

Energy poverty, health and education outcomes: Evidence from the developing world

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ABSTRACT

This study critically examines the effect of energy poverty on health and education outcomes for 50 developing countries in the period 1990–2017. We construct an aggregate energy development index and empirically test if the effect on development outcomes is conditioned by thresholds determined by the degree of poverty and income per capita. Our empirical results show that lower energy poverty is associated with higher health and education outcomes. However, we find that access to electricity has a more substantial positive effect on development outcomes than energy use. The threshold regression estimates show that there is a greater effect of energy development index on life expectancy rates where the poverty headcount ratio is high. Conversely, energy development index has a greater effect on infant mortality rates where the poverty headcount ratio is low, or income per capita is high. We do not find a threshold effect of poverty and income per capita in the education sector. These findings have strong policy implications for developing countries, showing that a lack of access to electricity remains a critical obstacle for achieving greater economic development, and poverty plays a crucial role in shaping these countries' health outcomes.

1. Introduction

The sustainable development goals (SDGs) set by the United Nations in 2015 aim at putting an end to all forms of poverty worldwide, including energy poverty, by 2030. Alleviation of energy poverty relies on three important pillars: improving electricity access and energy access for poor households towards cleaner and safer cooking fuels, efficiency in electricity use, and greater use of renewable energy (Biol, 2007). According to the World Energy Outlook 2020 report, about 770 million people (10% of the world population) had no access to electricity and about 2.6 billion people (34% of the world population) lacked access to clean fuel in 2019 (IEA, 2020). The recent outbreak of the COVID-19 pandemic further aggravates this problem. The United Nations estimates that the share of the world population experiencing extreme poverty will increase by another 71 million in 2020 due to the pandemic.¹ The current situation on energy poverty is complex in many developing countries because of the COVID-19 outbreak. The pandemic has brought a reversal of fortune for many sub-Saharan African countries that previously had electricity access, leaving an additional 2% (an

increase of 13 million) people without access to electricity in 2020 (IEA, 2020). Due to subsequent economic lockdowns and greater confinement and restrictions, there is an increased demand from residents to use their basic household appliances. This exerts additional pressure on the electricity grid. At the same time, energy-poor people are vulnerable to greater health risks from social and physical distancing, including mental illnesses. With higher job losses and lower investments due to the pandemic, people worldwide are struggling to meet their basic needs for survival, including electricity and other forms of energy (Mastropietro et al., 2020).

This paper contributes to the energy poverty literature by examining the impact of lack of access to and usage of electricity and other types of energy on health and education outcomes, for 50 developing countries in the period 1990–2017. Existing studies on energy poverty related to these outcomes in developing countries are typically limited to the household-level data (Ahmad et al., 2014; Grimm et al., 2015; Oum, 2019; Zhang et al., 2019). International evidence using a panel data approach is virtually absent in the extant literature. Our paper fills this gap and provides a fresh perspective on common deterring channels

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¹ Link: <https://www.un.org/development/desa/en/news/sustainable/sustainable-development-goals-report-2020.html>

related to the energy scarcity that developing countries experience worldwide. While economies can adopt their distinct energy-related strategies at a country level, international cooperation and joint strategies would help better capacity development, financing, and implementing best practices from the frontier markets (AGECC, 2010). Thus, findings from an international cross-country study like this will help poor nations develop a set of coordinated strategies that are more economically viable and achievable in the long run.

By considering two different development outcomes, we capture the heterogeneous effect of energy poverty on whether the effect is more critical for specific development outcomes than others. As an economy travels through different stages of development, the effect of energy poverty on education compared to health outcomes could be very different. For example, Deaton (2003) suggests that general poverty and health can influence each other; however, there is stronger evidence that with the redistribution of income from richer to poorer sections of the population, the improvement in the health status of the poor is more significant than the decline in the health status of the rich. Consequently, due to greater redistribution of income, if the number of poor people in a country is higher than the number of rich, the national average health status improves. Hence, reducing energy poverty could be more effective for health outcomes when the general poverty rate is lowered to a certain threshold level in developing countries. However, this hypothesis requires rigorous empirical testing before any meaningful conclusion can be reached. Of particular interest, we test whether the influence of energy poverty on health and education outcomes is conditioned by the degrees of poverty and income per capita in an economy.

Lack of access to electricity, clean fuels and inefficient energy use are major components of energy poverty and their roles in economic underdevelopment are already well established in the literature (Bonan et al., 2017; Cabraal et al., 2005). Nonetheless, the relationship between various components of energy poverty may not remain the same as an economy travels through different stages of economic development (Toman and Jemelkova, 2003). In impoverished economies with average per capita income levels of less than \$300 per year, electricity is almost absent, and more than 90% of the population depends on traditional means of cooking, such as wood and cow dung (Barnes and Toman, 2006). These economies typically experience high poverty rates, underdeveloped health and education sectors characterised by high infant mortality rates, lower life expectancy rates and lower human capital accumulation at all levels. With technological progress and an increased standard of living, the transition to modern energy occurs. Then, electric lights, clean cooking and other electronic appliances become more common among households. Therefore, at a very low level of economic progress, when poverty is high, and access to electricity is almost absent, energy usage is involuntarily low. However, with near-universal access, electricity usage and modern clean fuels are expected to have a higher impact on economic development. Hence, from a policy perspective it is important to examine the relative importance of aggregate energy progression and its dimensions separately in terms of economic development outcomes at different threshold levels of poverty and income. We test their relative importance by using a composite energy index and its two main dimensions, namely access to electricity and per capita energy use, respectively.

As an additional feature of this study, following the IEA (2010) report on energy poverty and UNDP's method of constructing a human development index, we construct an aggregate energy development index (EDI) for all 50 economies on an annual basis. This multidimensional aggregate index is based on four key indicators of electricity access, usage, renewable energy use and total energy consumption. This aggregate composite index will help policymakers track energy development overtime at a global level. Moreover, the composite index can be utilised to rank countries and identify nations that need energy-related financing, particularly to attain energy access and energy efficiency. International organisations, such as the World Bank and the United

Nations, can use this composite index of energy development to channel foreign aid and/or loan disbursements to the most effective sectors, reduce the strain on their funding resources and to control energy transformation systems in developing countries. We utilise both the aggregate index and its two main dimensions, viz. access to electricity and energy usage, separately, to provide a robust relationship between energy poverty and health and education outcomes.

We adopt several empirical strategies to examine the relationship between energy poverty and economic development: a fixed effect panel regression technique, a fixed effect two-stage least squares (FE-2SLS) strategy, the Lewbel (2012) method of internally generated instruments-based estimations and a threshold regression technique proposed by Hansen (1999). In the FE-2SLS strategy, we use night-time light data captured by satellite images as an instrument for energy poverty. Many studies consider night-time satellite images to measure electrification rates in developing countries (Doll and Pachauri, 2010; Dugoua et al., 2018; Elvidge et al., 2011). Hence, the stronger the brightness level of night-time light detected from outer space for a country in a given year, the higher the electrification rate and energy use in that country. Specifically, night light data is collected exogenously by the Defence Meteorological Satellite Program to monitor moonlight clouds and map the permanent light network on the earth's surface.² The measure can capture all cumulative channels of electricity access and energy uses. It is also highly unlikely that other unobserved factors can influence health or education outcomes through the channel of night light. To alter the satellite images of a country from outer space, contributions from a single factor must be colossal. On the other hand, night-time light captured using satellite images can influence certain development outcomes only through the channels of greater access to and usage of electricity and/or energy.

However, as a limitation to the FE-2SLS identification strategy, the night-time light data can also measure national and sub-national levels of wealth and urbanization, and are often used in the literature to proxy for aggregate income and population densities (Doll et al., 2006; Keola et al., 2015; Mellander et al., 2015). Although the variable empirically satisfies both the exogeneity and relevancy conditions of a valid instrument, there could be a weak relationship between night-time data and health and education outcomes through the channel of aggregate income. Thus, as an additional robustness check we employ the technique proposed by Lewbel (2012), which uses an identification strategy of constructing an internal instrument based on the presence of heteroskedasticity in the data. The internally generated instruments can be combined with the external instruments to increase the reliability of estimates when concern exists that the external instrument might be weak (Mishra and Smyth, 2015).

Using an aggregate EDI, we find strong evidence that higher energy development is associated with better health and education outcomes in developing countries. Nevertheless, the positive effect is not uniform across the two dimensions of energy development. One standard deviation (SD) increase in the EDI increases life expectancy rates between 0.764 and 0.783 SD and decreases infant mortality rates between -4.957 and -5.222 SD after controlling for endogeneity. Among the two main dimensions of the EDI, access to electricity has a stronger effect on health outcomes than energy use. Similarly, in the education sector, we find that one SD increase in the EDI increases progression to secondary school between 3.298 and 3.719 SD and increases average

² We use the yearly night time luminosity data for the countries included in our sample from the website <https://datainspace.org>. This dataset is compiled by Hodler and Raschky (2014) using the original night-time data from National Oceanic and Atmospheric Administration (2014) Version 4 DMSP-OLS Night-time Lights Time Series dataset and the GADM database of Global Administrative Areas. For more information about night time luminosity, dataset see: <https://ngdc.noaa.gov/eog/dmsp/downloadV4composites.html>. For more information about GADM database, please visit <http://www.gadm.org/>

years of schooling between 4.177 and 5.410 SD after controlling for endogeneity. The threshold regression estimates show that the marginal gains from lower energy poverty are greater for life expectancy rates where the threshold levels of poverty headcount ratio are high. Conversely, where the poverty headcount ratio is low, the effect from higher energy development is stronger and associated with a lower infant mortality rate. But as the threshold value of the poverty headcount ratio rises, the negative effect from energy development weakens. A similar effect is observed when income per capita is the threshold variable. As income per capita level increases, the negative effect of higher energy development is stronger for infant mortality rates. Thus, higher poverty level or lower income per capita weakens the benefits of energy development in reducing infant mortality rates. A similar outcome, however, is not observed in the education sector. Although energy poverty significantly influences education outcomes, we do not find a significant threshold effect of poverty and income per capita in the education sector.

The paper is organised as follows. Section 2 provides an overview of the literature showing the linkages between energy poverty with each of the development outcomes. In Section 3, we construct a multidimensional EDI for all 50 countries on an annual basis and present a broad outlook of energy poverty for these countries in the period 1990–2017. This section also discusses the measurements of the main variables and presents the empirical methodology. Section 4 discusses the empirical results. Finally, Section 5 concludes this discussion and provides future policy directions.

2. Literature review

Recent evidence suggests that higher energy poverty and fuel poverty lowers subjective wellbeing of households in poor as well as richer countries (Awaworyi Churchill et al., 2020; Biermann, 2016; Rodriguez-Alvarez et al., 2019; Thomson et al., 2017b). While energy poverty is defined as lack of access or inefficient use of modern energy services, fuel poverty refers to fuel debt or lower affordability of fuel costs required by households to maintain a comfortable living in diverse climatic conditions (Li et al., 2014; Thomson et al., 2016). Furthermore, both health and education outcomes are fundamental to subjective wellbeing and for the productivity of nations. Although very little attention has been given to investigating its possible relationship with energy poverty at the international level, evidence suggests that various forms of energy poverty influence health and education outcomes significantly at the individual and regional levels (Ahmad et al., 2014; Bonan et al., 2017; Grimm et al., 2015).

2.1. Energy poverty and health

Energy consumption by the residential sector, together with the commercial sector, positively influence health outcomes. Productive use of energy by residential households leads to an increased standard of living through higher demand for home lighting, cooking, television, refrigeration, drinking water, hot water, and other home appliances. However, indirectly, energy consumption improves the health outcomes of individuals through clean cooking fuels, availability of safe drinking water, better communication about health services through television and radios, less respiratory and/or cardiovascular diseases and premature deaths from exposure to indoor air pollution (Bonan et al., 2017; Cabraal et al., 2005; Gonzalez, 2015). At a commercial level, greater energy use facilitates better health care services, such as improved childbirth, proper lighting during medical procedures, better storage of vaccines and operation of medical equipment, and telecommunications in health clinics (Cabraal et al., 2005). Therefore, society will experience lower child mortality, maternal mortality, and higher life expectancy in both rural and urban areas.

A large body of empirical research supports the fact that higher energy poverty is associated with inferior health outcomes. The negative

effect on societies is perilous to future generations, as the physical health of children is particularly impacted. Due to higher fuel poverty, children are more prone to respiratory diseases and general illnesses (Liddell and Morris, 2010). The mental health of adults and caregivers is also affected as a result. In the absence of proper heating and insulation, cold and damp housing conditions exacerbate physical and mental health conditions (Liddell and Guiney, 2015; Tod and Thomson, 2017). Using nationally representative data for India, Ahmad et al., (2014) find that electricity availability significantly decreases illnesses, such as fever, cough and diarrhea, among households including very young children. Similarly, Phoumin and Kimura (2019) show that Cambodia's energy-poor households are more prone to respiratory diseases and end up spending more on hospital and medical bills.

The poor health status of energy-poor households is also prevalent among developed countries (Wilkinson et al., 2007). Coincidentally, energy poor households with lower health outcomes are prevalent in societies with relatively less income inequality and higher income, such as Sweden and Slovenia (Thomson et al., 2017b). Likewise, in France, which experiences extreme cold in winter, energy unaffordability, thermal discomfort and insufficient heating are widespread among residents (Kahouli, 2020). The rising healthcare costs put additional pressure on residents to maintain a comfortable standard of living. In the twenty-first century, rising health expenditure is also one of the greatest challenges for governments in rich countries to keep their fiscal budgets under control (Hall and Jones, 2007). A potential channel to control this increasing public expenditure on healthcare is to reduce the fuel poverty of residents in colder climates by making energy more affordable (Kahouli, 2020). But energy poverty is also present in developed countries outside Europe with warmer temperatures, such as in Australia, which is well known for its severe bush fires, heat waves in summer and low temperatures in winter. Using a multi-dimensional health measure, Awaworyi Churchill and Smyth (2021) find that energy poverty lowers general health outcomes of energy poor adult populations in Australia. Thus, by protecting those who are most vulnerable, general health conditions of a country can be significantly improved.

The adverse health outcomes due to indoor air pollutions are compounded in less developed countries, particularly in the rural areas with lower access to electricity and bad living conditions (González-Eguino, 2015; Oum, 2019). However, lower affordability of clean fuel and maintaining a good indoor temperature is not just restricted to rural households. In China, the health status of urban residents is negatively affected due to low quality indoor heating systems, choice of fuel, and lower affordability of building materials leading to weak insulation and bad ventilation (Wei and Liao, 2018a). Using detailed household-level survey data and a multidimensional measure of energy poverty, Zhang et al. (2019) find a significant negative impact of energy poverty on health in China. Whilst the above evidence is mostly either for a single country or for a small group of countries in a single year, we hypothesize that higher energy poverty is negatively associated with better health outcomes in developing countries.

2.2. Energy poverty and education

Education and health outcomes are intertwined and can influence each other at an aggregate level. While a healthier society delivers a conducive environment for better educational attainment and on-job training, people with higher education attainment are expected to stay happier, healthier and live longer than their less-educated equals (Zajacova and Lawrence, 2018). Hence, given the role of energy poverty in influencing health outcomes, impacts on education outcomes cannot be ignored. For instance, access to modern energy services provides greater flexibility for working adults to study at night and helps to reallocate the time of females from domestic work to higher education (Toman and Jemelkova, 2003). If a large section of society suffers from extreme poverty, the negative effect is higher for families with young children, particularly girls, who spend considerable time collecting

firewood and traditional cooking fuels instead of attending schools (Cabral et al., 2005; Khandker et al., 2014). Also, access to electricity and heating systems is essential for schools to provide education services in a colder climate. Evidence suggests that with each extra year of education and with access to electricity, the positive effect of electricity has a greater impact on household income (Cabral et al., 2005). Other benefits from better access to modern energy services may include proper lighting conditions inside classrooms, better working conditions for educators and caregivers, opportunities to run practical classes in laboratory-based courses and the operation of educational equipment, such as computers, photocopiers, televisions and projectors.

There is considerable empirical evidence at a micro-level showing that lower energy poverty is associated with higher education outcomes. Using household-level survey data in India and a multidimensional energy poverty index, Acharya and Sadath (2019) find that education plays a crucial role in impeding the spread of energy poverty. Further, evidence suggests that children with electricity spend more time studying than similar children without electricity (Khandker et al., 2014). The study also finds the positive effect on educational attainment is stronger for girls than boys, and thus improves average schooling in rural India. Similarly, electrification has a significant positive effect on education and employment outcomes in Brazil (Lipscomb et al., 2013), in sub-Saharan Africa (Bernard, 2012), in India (Ahmad et al., 2014) and in Lao PR (Oum, 2019). In a recent study, Zhang et al. (2021) show that academic scores in Chinese and maths subjects serve as potential channels through which energy poverty influences the subjective well-being of children in China. Despite a handful of studies that find no positive effect of lower energy poverty on education outcomes (Burlig and Preonas, 2016; Squires, 2015), evidence favouring better education outcomes is stronger. However, to this end we find no empirical work at the international level analysing the energy poverty-education nexus using a panel framework. Thus, based on the above evidence, we hypothesize that higher energy poverty is negatively associated with better education outcomes in developing countries.

2.3. General poverty and income as conditional variables

There is long-standing literature that shows the importance of energy poverty in human development, particularly in reducing general poverty levels (Birol, 2007; Dias et al., 2006; Wei and Liao, 2018b). Moreover, there is a strong association between multidimensional energy poverty index and income level for households in developing countries (Mendoza et al., 2019). The greatest benefits of lowering energy poverty levels are discernible in the agricultural sector by means of increased use of modern machines substituting manual labour, farm equipment, better irrigation systems and access to clean cooking fuels (Cabral et al., 2005). Once the agriculture sector is more productive, valuable resources can be transferred to other sectors, such as industries and services, which further lowers the overall poverty level and increases income. However, the benefits from lower energy poverty level can be extended beyond the agricultural sector by reducing barriers to affordable energy services for everyone, developing non-farm businesses, generating higher income, employment, and better health and education outcomes (Barnes and Toman, 2006). Besides, there is a burgeoning body of research demonstrating the association of economic progress with better education and health outcomes (van der Berg, 2008; Bergh et al., 2016; Deaton, 2007; Engle and Black, 2008; Marmot, 2005; Pickett and Wilkinson, 2015).

Since energy poverty is an essential component of general poverty, we propose the hypothesis that the relationship between energy poverty and development outcomes is conditional on the general poverty level. Energy poverty is expected to significantly affect developmental outcomes when the general poverty level is high. At the same time, when general poverty is fully alleviated in an economy, we can expect that the effect of energy poverty is negligible. However, during transition periods from higher to lower levels of general poverty, poorer sections of the

population are expected to have greater access to energy services and thus improved education and health outcomes. Given these inter-related linkages, an important contribution of this study is to examine the above hypothesis with the aim of general poverty level playing a conditioning role in the relationship between energy poverty and education and health outcomes. Further, since the poverty head count ratio in a society demonstrates the current living standards and an affordability indicator of basic goods and services, we consider income per capita as an additional threshold variable to test the relationship between energy poverty and education and health outcomes.

3. Data and empirical methodology

3.1. Data

We initially start with 72 developing countries from four different regions, namely Africa, Asia, Europe and Latin America. This is based on data availability of health and education outcomes and the other control variables used in the empirical estimations. These variables are detailed in the next two sections. The definitions and sources of our key variables are provided in Appendix 1. However, energy indicators are not available for 20 countries in our initial sample, particularly in the African region. We find either many missing observations or no data on energy use and electric power consumption for countries, such as Bhutan, Burkina Faso, Burundi, Cape Verde, Central African Republic, Chad, Gambia, Guinea, Guinea-Bissau, Lesotho, Liberia, Madagascar, Malawi, Mali, Mauritania, Rwanda, Sierra Leone, Seychelles, Swaziland and Uganda. Thus, after excluding these countries, we have 52 countries in our sample. Further, two more countries, Angola and Cambodia, were excluded due to missing observations on poverty headcount ratio, which is used as a threshold variable in our empirical estimates. Thus, our final sample consists of a balanced panel of 1400 observations from 50 countries over 28 years on an annual basis. The list of countries is presented in Appendix 2. We use satellite night-time light data compiled by Hodler and Raschky (2014) as an instrument for the EDI in our empirical estimations. The night-time light data is only available from 1990 onwards. To remain consistent with our identification strategy, we take 1990 as our starting year and extend our period of analysis from 1990 to 2017.

3.2. Energy development index

Energy poverty is broadly defined as the lack of access to and use of modern energy services (Acharya and Sadath, 2019; Thomson et al., 2016). However, the definition of energy poverty varies widely among researchers and practitioners. This is based on whether there is a direct impact on a household's welfare as a result of lack of access to modern energy or an indirect but related impact coming from energy related attainments (Villalobos et al., 2021). The first type of energy poverty is closer to the definition of fuel poverty at an individual level and concerned with the effect of lack of sufficient energy access on a household's welfare or subjective wellbeing (Awaworyi Churchill and Smyth, 2021, 2020; Hills, 2012; Thomson et al., 2017a, 2017b). In this measure, a household's subjective wellbeing is lowered due to energy deprivation and bad heating conditions, particularly during cold winter months. Although this is a comprehensive tool to capture a household's utility of energy access, the measure often suffers from self-reporting biases and survey inconsistencies (Herrero, 2017). On the other hand, energy poverty can be measured objectively by the proportion of people who spend a relatively higher share of their household income on energy bills and thus become more economically vulnerable, particularly if they come from lower socio-economic income groups in developing countries (Boardman, 1991; Healy and Clinch, 2004; Hills, 2012; Tod and Thomson, 2017). However, these measures can also suffer from sampling selection biases, causing overestimation or underestimation of the energy poverty rates (Herrero, 2017). Instead, a multidimensional

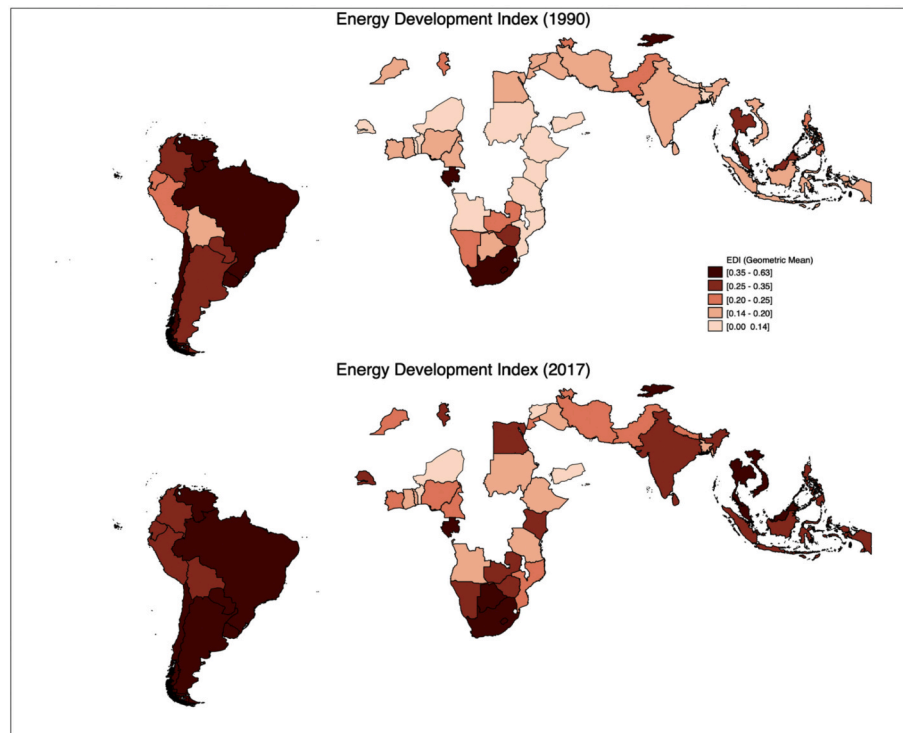


Fig. 1. Map of energy Development Index (EDI) for the period 1990–2017.
Source: World Development Indicators.

energy poverty index, which captures both objective and subjective measures, is less prone to sampling selection bias and survey-related errors (Nussbaumer et al., 2012; Villalobos et al., 2021). At an individual level, a number of studies adopt this methodology and create a multidimensional index of energy poverty for both the developing (Mendoza et al., 2019; Olawumi Israel-Akinbo et al., 2018; Sadath and Acharya, 2017; Zhang et al., 2019) and developed countries (Awaworyi Churchill and Smyth, 2021, 2020; Okushima, 2017; Thomson et al., 2017b, 2017a).

In comparison to the micro-level studies, there is no comprehensive multidimensional measure at an international level, which can be used to track and compare the transition of modern energy use in countries, including renewable energy. Similar to the micro-level studies in developing countries, an aggregate level multidimensional index is expected to capture both access to basic electricity and total energy use, including clean energy. Moreover, at an aggregate level it is important to capture both residential and commercial sectors since higher access to and modern energy use at the commercial level would imply that the country has a greater coverage of energy access and has adopted a greater number of policies related to modern energy transition. We construct a multidimensional energy development index (EDI) similar to that of the IEA (2010) report on energy poverty. This index is based on four indicators comprising both the residential and commercial sectors. They are: 1) total primary energy use per capita (measured in kilograms of oil equivalent); 2) electric power consumption per capita (kWh); 3) renewable energy consumption (% of total final energy consumption); and 4) access to electricity (% of the population). The definitions and

sources of the EDI components are listed in Appendix 1. The only closest study similar to our method of EDI construction is the energy development index constructed by the International Energy Association (IEA, 2010). However, unlike the IEA's method of using an arithmetic average of all four indicators, we follow the UNDP's method of constructing the human development index and take the geometric mean of all four indicators of the 50 economies on an annual basis in the period 1990–2017. This is because indicators are disproportionately weighted in arithmetic mean when data on an energy indicator is missing for a country compared to another country where data on all four energy indicators are available.

All four indicators of the EDI represent important dimensions of energy poverty. Higher primary energy use per capita refers to the use of primary energy before transition to other types of fuels, such as electricity and refined petroleum products. Renewable energy consumption is the share of renewable energy in total final energy consumption. Thus, higher energy use refers to higher consumption of all types of energy, which would be important for commercial and industrial growth and economic growth of transitional economies (Wolfram et al., 2012). On the other hand, renewable energy use would indicate the extent to which the economy is consuming modern, clean energy. Similarly, access to electricity and electric power consumption per capita demonstrate access to and usage of electricity in each country.

At first, each indicator is standardised to express the values between zero and one using the following formula:

$$Indicator_{ijt} = \frac{\text{actual value of country } j \text{ in year } t - \text{minimum value of country } j \text{ in year } t}{\text{maximum value of country } j \text{ in year } t - \text{minimum value of country } j \text{ in year } t} \quad (1)$$

After calculating the standardised value of all four indicators, the geometric mean is applied in the second step to combine all four indicators into a single index. We repeat the same process to construct the EDI for each country on an annual basis. Thus, the higher the value of the EDI of a country in a given year, the lower the energy poverty level.

Fig. 1 plots the EDI for all countries in our sample in 1990 and 2017. The differences between 1990 (first year of sample period) and 2017 (final year of sample period) demonstrate economic progress of countries in lowering energy poverty over time. While lighter shades of the EDI show lower values on the scale and refer to countries with lower levels of energy development, darker shades show higher values on the scale and refer to countries that experience higher levels of energy development. The lowest values of the EDI scores range between 0.00 and 0.14, and the highest values range from 0.35 to 0.63. Between 1990 and 2017, we find that many countries in our sample significantly moved up the scale of EDI scores. For example, in Asia and the Middle East, India, Sri Lanka, Iran, Thailand, Indonesia, and Vietnam all moved up along the scale in 2017 compared to 1990 values. We find similar

examples in Latin America but more prominently in African countries, such as Angola, Morocco, Nigeria, Cameroon, Namibia, Botswana, Kenya, Mozambique and others. Overall, the evidence suggests that many countries in our sample demonstrate significant economic progress in the energy sector.

3.3. Health and education outcomes

We follow the extant literature to measure health and education outcomes. We use ‘life expectancy at birth’ and ‘infant mortality rate’ to proxy health outcomes and ‘progression to secondary school’ and ‘average years of schooling’ to proxy education outcomes. As noted, all definitions and corresponding data sources of key variables are provided in Appendix 1. Infant mortality and life expectancy rates are considered as the principal health indicators at the country level from developmental policy perspectives (Wang, 2002). A number of studies have considered life expectancy and infant mortality rates to measure health outcomes,

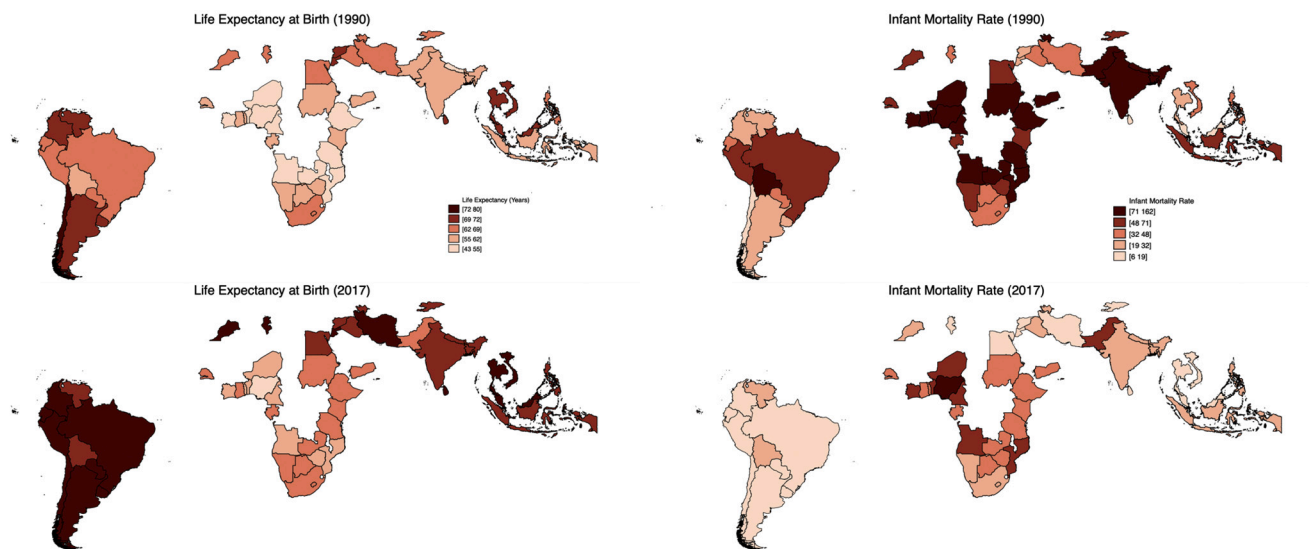


Fig. 2. Map of health variables (Life Expectancy at birth & Infant Mortality) for the period 1990–2017. Source: World Development Indicators.

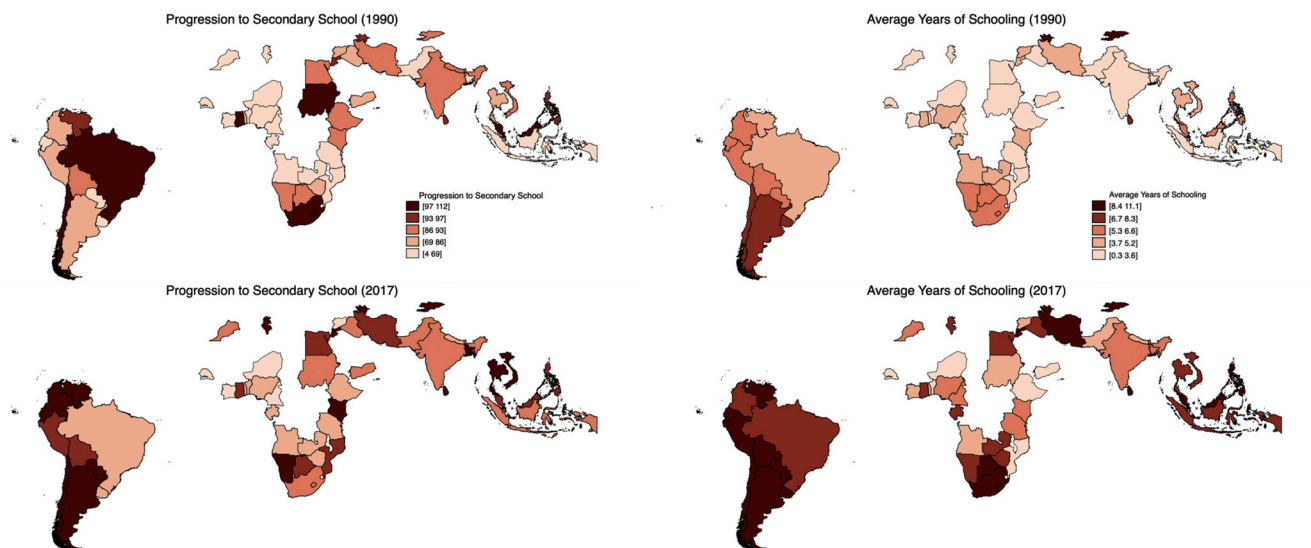


Fig. 3. Map of education variables (Progression to secondary school & Average years of schooling) for the period 1990–2017. Source: World Development Indicators.

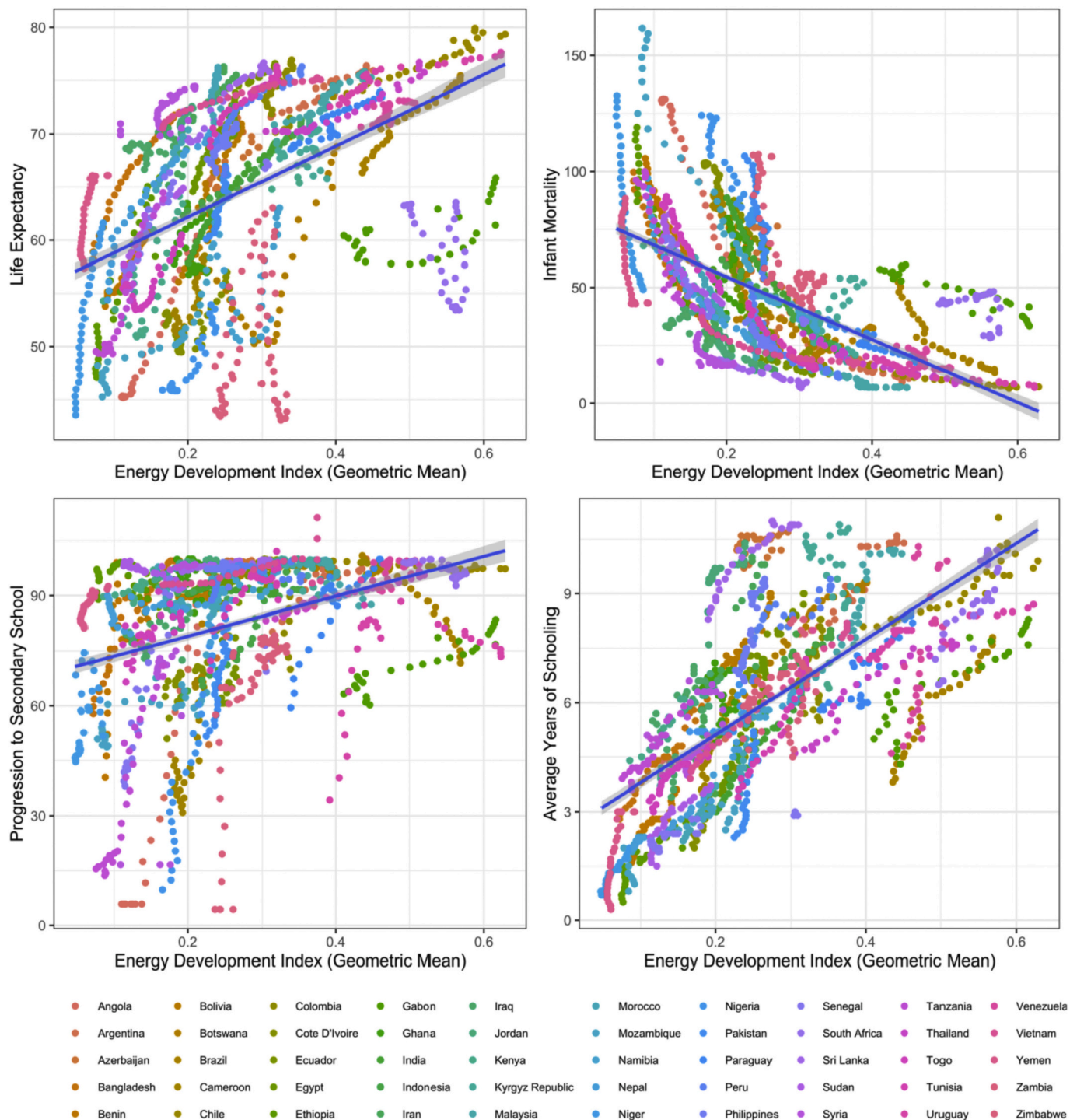


Fig. 4. Scatter plots showing the relationship between health and education outcomes with energy development index.

specifically when analysing energy poverty within a country. For instance, [Wilkinson et al. \(2007\)](#) note that to meet the UN millennium development goals on health, energy services should be provided globally and the effects can be captured by higher life expectancy and lower child mortality rates. Similarly, while analysing the effect of energy poverty on education in India, [Acharya and Sadath \(2019\)](#) use years of schooling for various states and territories in India to measure education outcomes.

[Fig. 2](#) plots life expectancy and infant mortality rates, respectively, and tracks the developments of countries in our sample between 1990 and 2017. The life expectancy rates are between 43 and 55 in the lowest

range and between 72 and 80 in their highest range. For infant mortality rates per 1000 births, the lowest range takes the values between 6 and 19 and the highest range between 71 and 162. Since lower infant mortality rates imply better health outcomes, lighter shades in the maps demonstrate better health conditions. Comparing life expectancy and infant mortality rates in [Fig. 2](#), we find that both of these health outcomes have improved over time. All countries in Asia and many countries in Africa and Latin America have progressed over time. While most countries in South America, Asia and Africa in our sample have moved from lighter shades to darker shades in life expectancy, the same countries have moved inversely for infant mortality rates. Overall,

evidence shows that most countries have significantly improved their health outcomes over time.

Fig. 3 plots the two education outcomes viz. progression to secondary schooling and average years of schooling, and tracks their development between 1990 and 2017. The lowest values of progression to secondary school range between 4 and 69 and the highest values range between 97 and 112. For average years of schooling, the lowest range is between 0.3 and 3.6, and the highest range is between 8.4 and 11.1. As expected, both figures demonstrate significant economic progress in the education sector over time. However, for some countries in Africa and Latin America, particularly countries such as Brazil, Sudan, Ghana, and South Africa, the progression rate to secondary school has fallen over time. Moreover, countries such as India and Algeria, the progression rates remained in the same range. The increase in average years of schooling is much more significant than progression rates. Except for some African countries, such as Algeria, Niger, Ethiopia, Mozambique and Yemen, all other countries increased their average years of schooling in 2017 compared to 1990. Moreover, compared with Fig. 1, we find a strong positive association between energy development and health and education outcomes between 1990 and 2017. For example, countries such as India, Thailand and Indonesia in the Asian region, Columbia, Brazil, Argentina, Chile and Uruguay in the Latin American region, and South Africa, Botswana, Zambia, Zimbabwe, Kenya and other countries in Africa, have all progressed in EDI scores in 2017 (Fig. 1). They have simultaneously achieved higher levels of life expectancy, lower infant mortality rates and higher average years of schooling compared to 1990 (Figs. 2 and 3).

The country-level summary statistics on energy, health and education variables are presented in Appendix 4. They provide a comprehensive view of how each country has performed in energy and development outcomes in the period 1990–2017. Despite the fact that all countries included in the analysis are developing, there is a considerable variation in their performance on energy, health and education outcomes. We find the countries that are ranked very low on the energy development index are the same countries with low performance on health and education outcomes, for example, Niger, Yemen, Ethiopia and Tanzania. Conversely, the countries that have performed well on the energy development index, such as Uruguay, Brazil, Chile, and South Africa, are also among the top performers on health and education outcomes in this period. Some notable exceptions to these observations are countries such as Azerbaijan, Zambia and Ecuador, which are in the middle of the distribution when it comes to the energy development index but are towards the bottom of the distribution on health and educational outcomes. The evidence closely resembles the heat map analysis presented earlier in Figs. 1, 2 and 3, respectively.

To test this relationship further, we present pairwise scatter plots between the EDI and health and education outcomes in Fig. 4. We find a strong positive association between the EDI and life expectancy, progression to secondary school and average years of schooling, respectively, and a strong negative association between the EDI and infant mortality rates. Thus, countries which have progressed relatively well on health and education outcomes since 1990 have also advanced well on energy progression. Comparing the linear trend lines, the positive trend lines for the EDI and life expectancy and average years of schooling are stronger than primary school progression rates. This shows that the countries are closely clustered for the EDI and life expectancy and the EDI and average years of schooling.

3.4. Other control variables

The choice of our control variables is based on the extant literature on the relationship between energy poverty and corresponding development outcomes. There is long-standing literature showing the importance of institutional quality in economic development, particularly in alleviating poverty and income inequality (Chong and Calderón, 2000a, 2000b). Higher corruption and political instability bring lower

trust among politicians and uncertainty in current government structures of developing countries. As a result of corrupt bureaucrats, funding gets misallocated along with the lower provision of public goods in the health and education sectors (Bousmah et al., 2016). Since access to electricity is a form of public goods, higher institutional quality has a positive relationship with access to electricity and renewable energy consumption (Ahlborg et al., 2015; Uzar, 2020). In our empirical estimations, we control for institutional quality of countries proxied by the rule of law. The variable reflects perceptions that to some extent citizens in that country have confidence in and abide by society's rules, particularly the quality of contract enforcement, property rights, the police, and the courts, as well as the likelihood of criminal violence. The indicator ranges from approximately −2.5 (weak governance) to 2.5 (strong governance). For data sources of all control variables, please refer to Appendix 2. Additionally, we control for the log of real GDP per capita and log of urban population growth in health and education regressions as they demonstrate rapid social change, which brings economic instability, changes in affordability and demand for essential goods and services (Bergh and Nilsson, 2010).

Many studies show that health and education outcomes influence each other (Zajacova and Lawrence, 2018). While healthy children can learn better and positively contribute to long-term economic growth, higher levels of education increase cognitive skills and overall knowledge in a society, and deliver longer life and better health outcomes (Behrman, 1996; Eide and Showalter, 2011). To control for the level of education, we include the log of average years of schooling in the population greater than 15 years of age in health regressions and log of life expectancy rates as a proxy for better health status in education regressions. Extant literature also shows that higher nutritional status provides better health quality and positively influences longevity (Bergh and Nilsson, 2010). We control nutritional status measured by the log of the prevalence of the undernourished share of the population in health regressions. Since our sample of countries ranges between very poor to developing and transitional economies, the log of the percentage of children between 12 and 23 months having access to immunization is added as an additional control in health regressions to control health-care quality in these countries. Similarly, we control for quality of education in the education regressions as measured by the pupil-teacher ratio in primary education, which captures better learning environments in schools (Dearden et al., 2002). Finally, as discussed earlier in section 2.3, we control the general poverty headcount ratio in all regressions, which significantly affects the health and education outcomes.

3.5. Baseline methodology

We utilise a panel fixed effect model to estimate energy poverty's effect on health and education outcomes. The model is specified as follows:

$$y_{jt} = \beta_0 + \beta_1 EDI_{jt} + \beta_2 C_{jt} + \mu_j + \varepsilon_{jt} \quad (2)$$

where y_{jt} takes various measures of health and education outcomes, j denotes the country ($j = 1, \dots, 50$) and t denotes the time ($t = 1990, \dots, 2017$). EDI_{jt} is the energy poverty index of country j in year t . C_{jt} represents the set of control variables. μ_j shows a vector of country dummies which captures unobserved country-specific fixed effect and ε_{jt} displays the idiosyncratic error term.

As noted earlier, to control for the endogeneity relationship between the EDI and development outcomes, we utilise a FE-2SLS identification strategy. Eq. (2) can be represented in the following two-stage equations.

$$y_{jt} = \beta_0 + \beta_1 \widehat{EDI}_{jt} + \beta_2 C_{jt} + \mu_j + \varepsilon'_{jt} \quad (3)$$

$$\widehat{EDI}_{jt} = \alpha_0 + \alpha_1 C_{jt} + \alpha_2 Nighttime_{jt} + \lambda_j + \nu_{jt} \quad (4)$$

Table 1
Summary statistics.

Variable	Mean	Median	Std. Dev.	Min	Max
Life expectancy at birth, total (years)	64.214	66.101	8.937	43.068	79.905
Infant mortality rate	46.320	40.104	29.476	6.404	161.702
Progression to secondary school	86.771	92.620	14.990	19.765	100.004
Average years of schooling	5.56	6.000	2.513	0.301	11.101
Energy Development Index	0.257	0.237	0.134	0.048	0.627
Access to electricity	65.876	73.604	31.248	2.279	100.004
Energy use	845.157	616.718	649.716	113.086	3428.559
Rule of law	−0.521	−0.579	0.644	−2.256	1.427
Real GDP per capita	3359.066	1909.01	3288.257	164.355	14,920.403
Poverty headcount ratio	23.403	15.645	22.382	0.001	86.202
Urban population growth	0.034	0.029	0.022	−0.063	0.108
Prevalence of undernourishment	13.957	11.901	9.788	2.501	67.502
Access to immunization, measles (% of children ages 12–23 months)	78.822	83.000	17.276	12.004	99.001
Pupil-teacher ratio, primary	31.588	29.057	11.792	11.022	67.337

Table 2
Fixed effects estimates-the impact of energy development index on life expectancy at birth, total (years).

Variables	Fixed effects estimates			2SLS estimates			Lewbel (2012) Internal and External		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Energy Development Index	0.431*** (11.33)			0.764*** (8.82)			0.783*** (9.593)		
Access to electricity		0.082*** (16.06)			0.248*** (7.02)			0.135*** (7.324)	
Energy use			0.043*** (7.45)			0.199*** (7.10)			0.063*** (5.623)
Rule of law	0.004 (0.78)	0.004 (0.83)	0.0052 (1.06)	0.004 (0.71)	0.003 (0.53)	0.010 (1.54)	0.004 (0.728)	0.004 (0.756)	0.006 (1.184)
Real GDP per capita	0.020 (1.05)	0.012 (0.66)	0.018 (0.95)	0.023 (1.14)	0.002 (0.08)	0.024 (0.98)	0.022 (1.161)	0.011 (0.615)	0.020 (1.064)
Poverty headcount ratio	−0.000 (−1.37)	0.000 (−0.11)	−0.001*** (−3.25)	0.000 (0.91)	−0.001*** (−3.90)	0.000 (0.87)	0.000 (1.081)	0.000** (2.031)	−0.000*** (−2.580)
Urban population growth	0.012*** (4.70)	0.012*** (5.15)	0.011*** (4.31)	0.010*** (3.99)	0.010*** (2.98)	0.002 (0.67)	0.010*** (4.045)	0.011*** (4.686)	0.010*** (3.853)
Prevalence of undernourishment	−0.011*** (−4.65)	−0.009*** (−3.94)	−0.015*** (−6.01)	−0.010*** (−3.99)	−0.002 (−0.46)	−0.022*** (−6.72)	−0.010*** (−4.037)	−0.007*** (−2.687)	−0.016*** (−6.410)
Average years of schooling	0.077*** (10.03)	0.076*** (10.80)	0.101*** (13.66)	0.045*** (4.06)	−0.014 (−0.65)	0.033** (2.19)	0.043*** (4.067)	0.047*** (3.981)	0.092*** (10.955)
Access to immunization, measles (% of children ages 12–23 months)	0.105*** (12.20)	0.080*** (9.67)	0.099*** (11.25)	0.113*** (12.49)	0.048*** (3.71)	0.112*** (10.03)	0.113*** (12.785)	0.070*** (7.701)	0.101*** (11.627)
Constant	3.533*** (88.58)	3.422*** (86.67)	3.366*** (62.19)	3.435*** (76.53)	2.976*** (28.99)	2.314*** (12.11)	−0.001 (−0.634)	−0.001 (−0.382)	−0.001 (−0.539)
Weak identification test (Cragg-Donald Wald F statistic)	—	—	—	342.483	52.457	95.728	47.170	14.383	57.687
Number of countries/Obs.	50/ 1336	50/ 1350	50/ 1350	50/ 1336	50/ 1350	50/ 1350	50/ 1336	50/ 1350	50/ 1350
Adj R-squared	0.39	0.58	0.46	0.51	0.2499	0.2657	0.520	0.553	0.522

Notes: The dependent variable is life expectancy at birth, total (years). Specifications (1) to (3) show country fixed effect estimates, specifications (4) to (6) show panel 2SLS estimates with country fixed effects and specifications (7) to (9) show Lewbel (2012) estimates with country fixed effects. Energy development index and its two main dimensions viz. access to electricity and total energy use have significantly positive effects on life expectancy under fixed effects and 2SLS estimations. The coefficients of our key independent variables (Energy Development Index, Access to electricity and Energy use) increase as we instrument them by night-time light. Specifications (4) to (9) show that the strength of our instrument measured by Cragg-Donald Wald F statistic is higher when we use overall energy development index than at dimension level. Access to electricity has higher effect on life expectancy than energy use. All variables are expressed in logarithmic form except Energy Development Index, Rule of law and Poverty headcount ratio. The coefficients of the control variables show the expected signs following the literature. ***, **, and * denote significant at the 1%, 5% and 10% level respectively. Heteroscedasticity and autocorrelation robust t-statistics are in parentheses. Our instruments are very strong as Cragg-Donald Wald F statistic is well above the conventional threshold (i.e., 10).

Key variables of interest are presented in bold.

where $Nighttime_{jt}$ shows the night-time light data in country j and at time t . The other variables have similar definitions as in Eq. (2). Eq. (4) is the first stage equation where we isolate the effect of the EDI. In the second stage, the fitted values of $\widehat{EDI}_{jt}$ derived from Eq. (4) are plugged into Eq. (3) to address the endogeneity issue between y_{jt} and EDI_{jt} . We use night-time light data captured by the satellite images as an instrument for the EDI. Night light data is used by the Defence Meteorological Satellite Program to capture weather elements and to map the light network on the earth's surface. Therefore, the stronger the brightness level of night-time light from outer space for a country in a given year, the higher electrification rate and energy use in that country (Doll and Pachauri,

2010; Dugoua et al., 2018; Elvidge et al., 2011). Night light satellite data can influence health and education outcomes only through the channels of higher electricity access and greater energy use, which signals lower energy poverty levels. To strengthen our empirical strategy, we also empirically test for the relevancy and exogeneity conditions of the instrument. We find the night light is positively correlated with the EDI and thus satisfies the relevancy condition of a valid instrument. Next, we utilise the critical values of the Cragg-Donald Wald F-statistic provided by Stock and Yogo (2005) to test if the instrument is weak in the presence of multiple endogenous regressors. The Cragg-Donald Wald F-statistic in the 2SLS estimations ranges between 50.6 and 376.9. As a rule of

Table 3

Fixed effects estimates-the impact of energy development index on infant mortality rate.

Variables	Fixed effects estimates			2SLS estimates			Lewbel (2012) Internal and External		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Energy Development Index	-3.150*** (-22.26)			-5.222*** (-15.43)			-4.957*** (-15.856)		
Access to electricity		-0.152*** (-6.65)			-1.695*** (-6.79)			-0.911*** (-8.422)	
Energy use			-0.308*** (-13.21)			-1.358*** (-9.41)			-0.310*** (-6.890)
Rule of law	0.084*** (4.71)	0.082*** (3.98)	0.074*** (3.72)	0.086*** (4.45)	0.087* (1.99)	0.045 (1.41)	0.085*** (4.602)	0.085*** (3.110)	0.074*** (3.821)
Real GDP per capita	-0.065 (-0.91)	-0.034 (-0.41)	-0.0534 (-0.68)	-0.079 (-1.03)	0.060 (0.35)	-0.089 (-0.71)	-0.072 (-1.001)	-0.001 (-0.012)	-0.058 (-0.780)
Poverty headcount ratio	0.002*** (3.62)	0.005*** (6.55)	0.005*** (6.87)	-0.000 (-0.36)	0.009*** (-3.41)	-0.001 (-0.42)	0.000 (0.052)	-0.002 (-1.579)	0.004*** (6.704)
Urban population growth	-0.029*** (-3.08)	-0.040*** (-3.69)	-0.025** (-2.39)	-0.020* (-1.97)	-0.016 (-0.68)	0.034* (1.88)	-0.021** (-2.166)	-0.028* (-1.925)	-0.024** (-2.352)
Prevalence of undernourishment	0.088*** (9.86)	0.093*** (8.98)	0.114*** (11.60)	0.080*** (8.29)	0.024 (0.96)	0.165*** (9.64)	0.080*** (8.698)	0.058*** (4.009)	0.114*** (11.618)
Average years of schooling	-0.528*** (-18.39)	-0.755*** (-23.97)	-0.703*** (-23.81)	-0.324*** (-7.55)	0.073 (0.50)	-0.244*** (-3.16)	-0.349*** (-8.693)	-0.346*** (-4.973)	-0.699*** (-20.921)
Access to immunization, measles (% of children ages 12–23 months)	-0.169*** (-5.25)	-0.070* (-1.87)	-0.124*** (-3.52)	-0.215*** (-6.10)	0.226** (2.47)	-0.211*** (-3.67)	-0.210*** (-6.201)	0.074 (1.391)	-0.127*** (-3.654)
Constant	5.703*** (38.39)	5.328*** (29.98)	6.865*** (31.66)	6.291*** (35.67)	9.428*** (12.97)	13.957*** (13.96)	-0.001 (-0.201)	-0.003 (-0.447)	-0.002 (-0.333)
Weak identification test (Cragg-Donald Wald F statistic)	–	–	–	328.186	50.636	92.244	47.170	14.383	57.687
Number of countries/Obs.	50/ 1350	50/ 1350	50/ 1350	50/1350	50/1350	50/1350	50/1350	50/1350	50/1350
Adj R-squared	0.56	0.56	0.56	0.70	0.51	0.23	0.718	0.380	0.694

Notes: The dependent variable is infant mortality rate. Specifications (1) to (3) show country fixed effect estimates, specifications (4) to (6) show panel 2SLS estimates with country fixed effects and specifications (7) to (9) show Lewbel (2012) estimates with country fixed effects and external instruments combined with internally generated instruments. Energy development index and its two main dimensions viz. access to electricity and total energy use have significantly negative effects on life expectancy under fixed effects and 2SLS estimations. The coefficients of our key independent variables (Energy Development Index, Access to electricity and Energy use) increase as we instrument them by night-time light. Specifications (4) to (9) show that the strength of our instrument measured by Cragg-Donald Wald F statistic is higher when we use overall energy development index than at dimension level. Access to electricity has higher effect on infant mortality than energy use under 2SLS estimation. All variables are expressed in logarithmic form except Energy Development Index, Rule of law and Poverty headcount ratio. The coefficients of the control variables show the expected signs following the literature. ***, **, and * denote significant at the 1%, 5% and 10% level respectively. Heteroscedasticity and autocorrelation robust t-statistics are in parentheses. Our instruments are very strong as Cragg-Donald Wald F statistic is well above the conventional threshold (i.e., 10). Key variables of interest are presented in bold.

thumb, if the test statistic is found to be greater than 10, the instrument is believed to satisfy the second stage of exclusion restrictions (Kaufmann et al., 2019). Thus, our instrument satisfies both the relevancy and the exogeneity conditions of a valid instrument.

Nonetheless, higher night-time light data can also measure economic progress of countries over time as a result of higher aggregate income (Doll et al., 2006; Keola et al., 2015; Mellander et al., 2015). Hence, there could be a weak relationship between night-time light data and health and education outcomes through the channel of aggregate income. As an additional robustness test, we follow Lewbel (2012) to generate external instruments combined with internally generated instruments. This technique is particularly useful for empirical models where other forms of identification are either weak or not found.³

3.6. Panel threshold regression

We also investigate the possibility of non-linearity in the relationship between the EDI and health and education outcomes using an endogenous threshold regression model proposed by Hansen (1999). Given the complex nature of the relationship between poverty and income levels with the health and education outcomes, it is possible that the regression function is not identical across all observations in the sample. Using the

³ The Stata module uses the command 'ivreg2h' to estimate Lewbel (2012). Nonetheless, the command 'ivreg2h' does not provide any explicit support for using panel data. We use the centering of variables approach suggested by Baum and Schaffer (2012) to remove country fixed effects before estimating the Lewbel model.

threshold regression approach, we entertain the idea that the regression relationship between the EDI and health and education variables can be divided into discrete classes depending upon an endogenously determined level of poverty or the country's per capita income at a point in time. In most simple terms, the threshold regression model can be described as follows (borrowing the notation used by Hansen (1999)).

In a balanced panel data model, the dependent variable of health or education is represented by y_{jt} . The threshold variable of poverty or per capita income is defined by q_{jt} . A k dimensional vector x_{jt} represents the independent variables. The k dimensional vector x_{jt} includes the key independent variable, EDI_{jt} , and other control variables (C_{jt}) as specified in Eq. (2) above. Notations of other variables are same as Eq. (3) above. The following structural equation can describe the threshold model.

$$y_{jt} = \beta_0 + \beta_1' x_{jt} I(q_{jt} \leq \gamma) + \beta_2' x_{jt} I(q_{jt} > \gamma) + \mu_j + \epsilon_{jt}' \quad (5)$$

$I(\cdot)$ is an indicator function that divides the observations into two regimes. The first regime consists of observations for which threshold variable q_{jt} is less than or equal to threshold value γ . The second regime consists of observations for which the threshold variable q_{jt} is greater than the threshold value γ .

Eq. (5) can also be expressed as the following set of equations.

$$y_{jt} = \begin{cases} \beta_0 + \beta_1' x_{jt} + \mu_j + \epsilon_{jt}', & q_{jt} \leq \gamma \\ \beta_0 + \beta_2' x_{jt} + \mu_j + \epsilon_{jt}', & q_{jt} > \gamma \end{cases} \quad (6)$$

As can be seen from Eq. (6) above, the identification would require the dependent variable and the threshold variable to be not time invariant. The regimes differ from each other by having different slopes; β_1 & β_2 for each regime.

Table 4

Fixed effects estimates-the impact of energy development index on progression to secondary school.

Variables	Fixed effects estimates			2SLS estimates			Lewbel (2012) Internal and External		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Energy Development Index	0.998*** (4.13)			3.719*** (7.00)			3.298*** (4.168)		
Access to electricity		0.141*** (4.18)			1.245*** (5.43)			0.543** (2.354)	
Energy use			0.016 (0.44)			1.030*** (5.72)			0.235** (2.511)
Rule of law	0.107*** (3.47)	0.107*** (3.77)	0.108*** (3.74)	0.106*** (3.54)	0.112*** (2.90)	0.134*** (3.65)	0.091* (1.839)	0.094* (1.877)	0.099** (1.992)
Real GDP per capita	0.169 (1.49)	0.156 (1.38)	0.161 (1.41)	0.191 (1.61)	0.123 (0.81)	0.226 (1.55)	0.540*** (2.816)	0.506*** (2.616)	0.529*** (2.770)
Poverty headcount ratio	0.001 (0.75)	0.001 (1.20)	−0.000 (−0.21)	0.003*** (3.02)	0.012*** (4.68)	0.002 (1.53)	0.003* (1.857)	0.005** (1.986)	0.001 (0.412)
Urban population growth	0.111*** (7.60)	0.108*** (7.46)	0.104*** (7.14)	0.129*** (8.29)	0.141*** (6.83)	0.101*** (5.48)	0.117*** (4.531)	0.111*** (4.209)	0.094*** (3.750)
Life expectancy	0.030*** (11.61)	0.029*** (10.68)	0.035*** (14.62)	0.015*** (4.18)	−0.021* (−1.95)	0.012** (2.38)	0.017*** (2.942)	0.010 (0.897)	0.029*** (6.605)
Pupil-teacher ratio, primary	0.006*** (4.52)	0.003*** (2.72)	0.004*** (3.23)	0.011*** (6.78)	−0.002 (−0.92)	0.016*** (6.14)	0.009*** (3.611)	0.001 (0.270)	0.006** (2.502)
Constant	2.421*** (14.08)	2.244*** (12.79)	2.308*** (8.87)	2.440*** (13.20)	1.014*** (3.01)	−3.068*** (−3.05)	0.001 (0.188)	0.002 (0.208)	0.003 (0.399)
Weak identification test (Cragg-Donald Wald F statistic)	–	–	–	376.886	53.103	87.855	75.176	13.565	145.845
Number of countries/Obs.	50/ 1350	50/ 1350	50/ 1350	50/ 1350	50/ 1350	50/ 1350	50/ 1350	50/ 1350	50/ 1350
Adj R-squared	0.49	0.49	0.48	0.15	0.35	0.22	0.184	0.167	0.191

Notes: The dependent variable is progression to secondary school. Specifications (1) to (3) show country fixed effect estimates, specifications (4) to (6) show panel 2SLS estimates with country fixed effects and specifications (7) to (9) show Lewbel (2012) estimates with country fixed effects and external instruments combined with internally generated instruments. Energy development index and its two main dimensions viz. access to electricity and total energy use have significantly positive effects on progression to secondary school. Under fixed effects and 2SLS estimations. The coefficients of our key independent variables (Energy Development Index, Access to electricity and Energy use) increase as we instrument them by night-time light. Specifications (4) to (9) show that the strength of our instrument measured by Cragg-Donald Wald F statistic is higher when we use overall energy development index than at dimension level. Access to electricity has higher effect on progression to secondary school than energy use. All variables are expressed in logarithmic form except Energy Development Index, Rule of law and Poverty headcount ratio. The coefficients of the control variables show the expected signs following the literature. ***, **, and * denote significant at the 1%, 5% and 10% level respectively. Heteroscedasticity and autocorrelation robust t-statistics are in parentheses. Our instruments are very strong as Cragg-Donald Wald F statistic is well above the conventional threshold (i.e., 10).

Key variables of interest are presented in bold.

Eq. (6) can be generalised to a multiple threshold model by assuming that the threshold variable γ takes more than one discrete value. For instance, a double threshold model, using the above notation can be written as follows:

$$y_{jt} = \beta_0 + \beta'_1 x_{jt} I(q_{jt} \leq \gamma_1) + \beta'_2 x_{jt} I(\gamma_1 < q_{jt} \leq \gamma_2) + \beta'_3 x_{jt} I(\gamma_2 < q_{jt}) + \mu_j + \varepsilon'_{jt} \quad (7)$$

The rationale behind using poverty and per capita income as conditional variables is that the EDI may have a different impact on health and education outcomes depending on the country's level of poverty or its per capita income as discussed in section 2. The poverty level in a country can be seen as a rough indicator of energy affordability, specifically to afford clean energy (Birol, 2007). In contrast, the income per capita level is a rough gauge of a country's capacity to collect tax revenues towards providing energy infrastructure (Metcalf, 2010). Hence, it is reasonable to model the critical demand and supply-side constraints as threshold variables in the EDI and health and education relationships. To estimate the endogenous threshold model, we start with testing for the presence of one significant threshold in the data. If we find one threshold model is significant, we test for the double threshold model and so on. We continue to test for an increasing number of thresholds until we reach a model with no significant thresholds. This is consistent with the approach proposed by Hansen (1999).⁴

⁴ We use the Stata program command 'xthreg' to test for the significance of thresholds and estimate the model. The link to codes for estimating Threshold regression model can be found on Professor Bruce Hansen's home page at the following link: https://www.ssc.wisc.edu/~bhansen/progs/joe_99.html

4. Empirical results

4.1. Summary statistics

Table 1 presents the summary statistics of the key variables. Life expectancy at birth ranges between 43.07 and 79.91 with a mean value of 64.21 and standard deviation of 8.94. The mean value is closer to the median value of 66.10 (50th percentile), which shows the series is uniformly distributed. On the other hand, infant mortality rates range between 6.40 and 161.70. The variation is significantly higher in infant mortality rates than in life expectancy rates with a standard deviation of 29.48. In the education sector, the mean and median values of progression from primary to secondary schooling are 86.77 and 92.62, respectively, which are considerably higher as the theoretical maximum can reach a value of 100. However, the mean value of the average years of schooling is 5.96, which is considerably lower than the maximum value of 11.10. The EDI ranges between 0.05 and 0.63 with a mean value of 0.26. The mean is also closer to the median value of 0.24, showing scope for improvement in energy access and use in developing countries.

While the percentage of the population which has access to electricity ranges from 2.28% to 100%, the mean value is only 65.88. The mean is also lower than the median value of 73.60, which shows the data is skewed to the left. Thus, there is a considerable percentage of the sample population which does not have access to electricity. The total energy use variable also shows a considerable variation in the sample with a minimum of 113.09 kWh and 3428.56 kWh and a standard deviation of 649.72. However, the mean value (845.16 kWh) is higher than the median value (616.72 kWh), which shows that the data is skewed to the right. The finding indicates that those who have access to electricity and other energy sources continue to use energy considerably. Among the control variables, the mean value of access to immunization is 78.82,

Table 5
Fixed effects estimates-the impact of energy development index on average years of schooling.

Variables	Fixed effects estimates			2SLS estimates			Lewbel (2012) Internal and External		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Energy Development Index	1.854*** (13.06)			5.410*** (14.89)			4.177*** (14.523)		
Access to electricity		0.158*** (7.61)			1.812*** (7.15)			1.104*** (8.676)	
Energy use			0.180*** (8.33)			1.498*** (8.90)			0.218*** (6.762)
Rule of law	−0.016 (−0.96)	−0.015 (−0.85)	−0.011 (−0.61)	−0.018 (−0.86)	−0.009 (−0.21)	0.024 (0.70)	−0.016 (−0.896)	−0.011 (−0.409)	−0.009 (−0.529)
Real GDP per capita	0.007 (0.10)	−0.014 (−0.19)	0.003 (0.04)	0.036 (0.45)	−0.062 (−0.37)	0.087 (0.64)	0.008 (0.122)	−0.042 (−0.391)	−0.010 (−0.152)
Poverty headcount ratio	−0.004*** (−7.70)	−0.004*** (−7.48)	−0.006*** (−10.08)	−0.001 (−1.28)	−0.011*** (4.16)	−0.003** (−2.57)	−0.002*** (−3.297)	0.005*** (3.170)	−0.005*** (−10.103)
Urban population growth	−0.070*** (−8.16)	−0.077*** (−8.73)	−0.083*** (−9.37)	−0.046*** (−4.35)	−0.028 (−1.24)	−0.089*** (−4.96)	−0.054*** (−5.796)	−0.049*** (−3.388)	−0.083*** (−9.544)
Life expectancy	0.024*** (15.42)	0.026*** (15.81)	0.029*** (20.19)	0.004* (1.74)	−0.049*** (−4.08)	−0.001 (−0.16)	0.011*** (5.276)	−0.017*** (−2.692)	0.028*** (18.642)
Pupil-teacher ratio, primary	0.005*** (6.94)	0.001 (1.48)	0.004*** (5.06)	0.012*** (10.81)	−0.007*** (−3.05)	0.019*** (7.93)	0.009*** (10.282)	−0.003** (−2.485)	0.004*** (5.429)
Constant	−0.660*** (−6.52)	−0.878*** (−8.17)	−1.691*** (−10.73)	−0.527*** (−4.26)	−2.671*** (−7.11)	−8.807*** (−9.40)	0.005 (1.152)	0.006 (0.930)	0.007* (1.781)
Weak identification test (Cragg-Donald Wald F statistic)	—	—	—	376.513	53.103	87.855	45.176	13.565	145.845
Number of countries/Obs.	50/1350	50/1350	50/1350	50/1350	50/1350	50/1350	50/1350	50/1350	50/1350
Adj R-squared	0.49	0.37	0.44	0.37	0.35	0.75	0.489	0.199	0.544

Notes: The dependent variable is average years of schooling. Specifications (1) to (3) show country fixed effect estimates, specifications (4) to (6) show panel 2SLS estimates with country fixed effects and specifications (7) to (9) show Lewbel (2012) estimates with external instruments combined with internally generated instruments. Energy development index and its two main dimensions viz. access to electricity and total energy use have significantly positive effects on average years of schooling under fixed effects and 2SLS estimations. The coefficients of our key independent variables (Energy Development Index, Access to electricity and Energy use) increase as we instrument them by night-time light. Specifications (4) to (9) show that the strength of our instrument measured by Cragg-Donald Wald F statistic is higher when we use overall energy development index than at dimension level. Access to electricity has higher effect on average years of schooling than energy use under 2SLS estimation. All variables are expressed in logarithmic form except Energy Development Index, Rule of law and Poverty headcount ratio. The coefficients of the control variables show the expected signs following the literature. ***, **, and * denote significant at the 1%, 5% and 10% level respectively. Heteroscedasticity and autocorrelation robust t-statistics are in parentheses. Our instruments are very strong as Cragg-Donald Wald F statistic is well above the conventional threshold (i.e., 10).

Key variables of interest are presented in bold.

but the median value is 83. This indicates the data is skewed to the left, and a more significant percentage of the population do not have access to immunization. The pairwise correlation among key variables is presented in Appendix 4. The correlation coefficients are within range, except a few, which show a value greater than 0.6. The variables which are found to be highly correlated are expected as per the extant literature. The summary statistics of all other control variables in Table 1 show that they are within a reasonable range of values.

4.2. Health outcomes and energy poverty

We consider two health outcome variables viz. life expectancy rates and infant mortality rates, to empirically test the relationship with energy poverty. These results are presented in Tables 2 and 3, respectively. Both measures are widely used in the literature as superior health outcomes, and by using an alternative measure, we check for robustness of results. Table 2 presents the effect of the EDI and its two main dimensions, viz. access to electricity and total energy use, on life expectancy at birth generated from panel fixed effects, 2SLS estimations and the Lewbel technique to control for endogeneity. While specifications (1) to (3) show the effects of using panel fixed effects, specifications (4) to (6) show the same for panel 2SLS estimations and specifications (7) to (9) show the results for the Lewbel method of estimations. Under all three empirical strategies, we find that the EDI significantly increases life expectancy at birth (specifications 1, 4 and 7). Moreover, the EDI has a higher positive effect on life expectancy at birth in the 2SLS (0.763) and Lewbel (0.783) estimates as compared to the fixed effects (0.430). The Cragg-Donald statistics in 2SLS estimations are well above the threshold value and thus show our instrument is valid and robust. Hence, controlling for endogeneity provides a higher and more robust

relationship between the EDI and life expectancy rate.

In specifications 2, 3, 5, 6, 8 and 9 of Table 2, we test the relationship of two main dimensions of the EDI, viz. access to electricity and total energy use, with life expectancy at birth. Our results show that both access to electricity and energy use positively influence life expectancy at birth. However, access to electricity has a higher positive effect compared to energy use in all three estimation techniques. While one SD increase in access to electricity increases life expectancy at birth by 0.082 SD in specification 2, 0.248 SD in specification 5 and 0.135 in specification 8, one SD increase in total energy use increases life expectancy at birth by 0.043 SD in specification 3, 0.199 SD in specification 6 and 0.063 in specification 9, respectively. Like the aggregate EDI, the coefficients of the two key dimensions under 2SLS estimates in specifications 5 and 6 are higher than the fixed effects and Lewbel estimates.

The coefficients of the control variables show the expected signs following the literature. Although the rule of law and real GDP per capita are not significant, the poverty headcount ratio is negative and significant in two out of 6 specifications. Thus, a higher poverty rate has a negative effect on life expectancy rate. Urban population growth is positive and significant, showing a positive effect of structural change in the economy. Higher urban population growth shows higher demand for goods and services and lower mortality rates, positively impacting life expectancy rates. Similarly, the prevalence of undernourishment has a negative effect and average years of schooling and access to immunization, measles (% of children ages 12–23 months) have positive and significant effects in nearly all specifications.

Next, in Table 3, we use an alternative health outcome – infant mortality rate using fixed effects, panel 2SLS and Lewbel estimations. We find that the EDI has a significantly negative effect on infant

Table 6
Threshold effect in the relationship between health and energy poverty.

Dependent Variable →	Thresholds in Poverty Headcount ratio		Thresholds in Income Per Capita	
	Life Expectancy at Birth	Infant Mortality Rate	Life Expectancy at Birth	Infant Mortality Rate
<i>Test for Threshold effect</i>				
Number of Significant Thresholds	2	1	0	2
F-Statistics	90.65	86.73	29.13	82.67
P-Value	0.070	0.036	0.730	0.103
<i>Threshold Estimates⁽¹⁾</i>				
$\hat{\gamma}_1$	31.40	5.80		7.38
	[31.04–31.56]	[5.30–6.03]		[7.37–7.39]
$\hat{\gamma}_2$	49.20			8.07
	[48.41–49.50]			[8.04–8.08]
<i>Impact of Energy Development Index⁽²⁾</i>				
$\hat{\beta}_1$	0.328***	−3.079***		−1.719***
	(0.035)	(0.137)		(0.161)
$\hat{\beta}_2$	0.498***	−2.664***		−2.433***
	(0.038)	(0.147)		(0.148)
$\hat{\beta}_3$	0.890***			−3.237***
	(0.046)			(0.130)
<i>Impact of covariates</i>				
Rule of law	0.004	0.093***		0.090***
	(0.004)	(0.017)		(0.017)
Real GDP per capita	0.017	−0.047		−0.037
	(0.018)	(0.069)		(0.066)
Poverty headcount ratio	−0.002***	0.002***		0.003***
	(0.000)	(0.001)		(0.001)
Urban population growth	0.009***	−0.028***		−0.021**
	(0.002)	(0.009)		(0.009)
Prevalence of undernourishment	−0.008***	0.077***		0.069***
	(0.002)	(0.009)		(0.008)
Average years of schooling	0.072***	−0.530***		−0.547***
	(0.007)	(0.028)		(0.026)
Access to immunization, measles (% of children ages 12–23 months)	0.090***	−0.202***		−0.152***
	(0.008)	(0.031)		(0.030)
Constant	3.636***	5.809***		5.597***
	(0.037)	(0.144)		(0.137)
Observations	1350	1350		1350
Number of Countries	50	50		50
R-squared	0.622	0.765		0.788

Notes: (1) The values in square brackets are 95% confidence intervals (2) The Standard errors in parentheses. (3) ***, **, and * denote significant at the 1%, 5% and 10% level respectively (4) For the models with 2 thresholds $\hat{\beta}_1$ is the coefficient of EDI_{it} when the threshold variable (poverty or per capita income) $q_{it} \leq \hat{\gamma}_1$; $\hat{\beta}_2$ is the coefficient of EDI_{it} when the threshold variable $\hat{\gamma}_1 < q_{it} \leq \hat{\gamma}_2$ and $\hat{\beta}_3$ is the coefficient of EDI_{it} when the threshold variable $q_{it} > \hat{\gamma}_2$. (5) For the models with 1 thresholds $\hat{\beta}_1$ is the coefficient of EDI_{it} when the threshold variable (poverty or per capita income) $q_{it} \leq \hat{\gamma}_1$; $\hat{\beta}_2$ is the coefficient of EDI_{it} when the threshold variable $q_{it} > \hat{\gamma}_1$. (6) The choice of appropriate threshold model for a specification was determined by the number of significant thresholds detected in a Model. (7) For the models with no significant threshold the results boil down to fixed effect model presented in Tables 2 & 3. Country fixed effects are included in all models.

mortality rate (see specifications 1, 4 and 7). One SD increase in the EDI decreases infant mortality rates by −3.150 SD in specification 1 (panel fixed effects), −5.222 SD in specification 4 (2SLS estimates) and −4.957 SD in specification 7 (Lewbel estimates). Similar to Table 2, we find that the coefficient in 2SLS estimates (specification 4) is higher than panel fixed effects (specification 1) and Lewbel estimates (specification 7). Moreover, access to electricity and energy use have significantly negative effects on infant mortality rates. Comparing the two key dimensions, we find access to electricity has a higher negative effect (−1.695 and −0.911) on infant mortality rate than total energy use (−1.359 and −0.310) after controlling for endogeneity. The coefficients of the control variables, such as the rule of law, urban population growth, the prevalence of undernourishment, average years of schooling and access to immunization, are all significant and show the expected signs.

Together the findings from Tables 2 and 3 show that the EDI has a significant relationship with life expectancy and infant mortality rates after controlling for a range of sector-specific structural and quality variables. Higher energy development or lower energy poverty is associated

with a higher level of life expectancy and lower infant mortality rates. The finding is critical for developing countries, particularly energy-poor countries which lack quality healthcare services. These may require improved energy access and use during medical procedures in childbirth, uninterrupted energy use during critical surgery, better storage of vaccines and the operation of medical equipment in health clinics (Cabraal et al., 2005). All of these factors lead to higher life expectancy and lower infant mortality in poor societies. Moreover, the coefficient of access to electricity is stronger than the total energy use in most specifications, particularly in the 2SLS and Lewbel estimates, which shows that access to electricity remains a stronger barrier in poorly developed countries. In the absence of electricity, living conditions are worse due to higher indoor air pollution and unhygienic cooking fuels, such as cow dung and firewood (Bonan et al., 2017; Cabraal et al., 2005; Gonzalez, 2015). Thus, by providing greater access to electricity, policymakers would reduce premature deaths resulting from cardiovascular diseases and improve longevity in men, women and children.

4.3. Education outcomes and energy poverty

In [Tables 4 and 5](#), we report the effect of the EDI on education outcomes, measured by progression to secondary school ([Table 4](#)) and average years of schooling ([Table 5](#)). Similar to earlier tables, specifications 1–3 show the results for fixed effects, specifications 4–6 show the same for panel 2SLS estimates and specifications 7–9 show the results for Lewbel estimates. Again, the results show that the EDI has a significantly positive effect on progression to secondary school for all three types of empirical strategies (see specifications 1, 4 and 7). One SD increase in the EDI increases progression to secondary school by 0.998 SD in specification 1, 3.719 SD in specification 4 and 3.298 SD in specification 7. We find that after controlling for endogeneity, the positive effect of energy on progression rates is stronger and significant at the 1% level.

In [Table 4](#), specifications 2, 3, 5, 6, 8 and 9 show the effect of access to electricity and energy use on progression to secondary school utilising fixed effects, panel 2SLS and Lewbel estimations. Total energy use is significant and associated with an increase in progression to secondary school only when we control for endogeneity (1.030 in specification 6 and 0.235 in specification 9); however, the coefficients are still lower than access to electricity (1.245 in specification 5 and 0.543 in specification 8). The control variables have expected signs in the bulk of the estimations. Rule of law is positive and significant in all specifications showing better institutional quality is critical in advancing education in developing countries. The coefficients of urban population growth, life-expectancy and pupil to teacher ratio in primary school are positive and significant. In the education regressions, urban population growth controls for any structural change in the economy over time and life expectancy rates control for health conditions in the economy. Finally, the pupil to teacher ratio in primary school captures the quality of schooling in developing countries. The findings show that all three control variables are positive and significant, bear the expected sign and are robust across most of the specifications.

We use an alternative education outcome, average years of schooling, in [Table 5](#). The results are very consistent with [Table 4](#). The EDI has a significantly positive effect on average years of schooling. The coefficient of the EDI is 1.854 in specification 1, 5.410 in specification 4, and 4.177 in specification 7. Thus, after controlling for endogeneity, the effect of energy development on average years of schooling is higher. Also, both access to electricity and energy use positively affect average years of schooling for fixed effects, 2SLS and Lewbel estimations. Consistent with previous findings, after controlling for endogeneity the coefficient of access to electricity is higher (1.812 in specification 5 and 1.104 in specification 8) than total energy use (1.498 in specification 6 and 0.218 in specification 9). The Cragg-Donald Wald statistics for 2SLS estimates in specifications (4) to (6) show that our instrument is valid and robust across all specifications. As before, the control variables show expected signs and are significant in the bulk of the estimations in [Table 5](#).

Overall, [Tables 4 and 5](#) show that the EDI is positively associated with educational outcomes in developing countries. While a large proportion of young children, particularly girls, in developing countries drop out of school to collect firewood or traditional cooking fuels ([Khandker et al., 2014](#)), our findings support the literature and show that higher access to electricity and clean cooking fuels lower school dropout rates and improves the progression of students from primary to secondary schools at an international level. Moreover, after controlling for endogeneity and similar to the health sector, the association between education outcomes and electricity access is higher than the total energy use. Since better access to electricity provides better learning conditions inside classrooms, better temperature control and better working conditions for educators and caregivers ([Cabral et al., 2005](#)), we find an increased effect on average years of schooling at an international level. The marginal gain from greater access to electricity in energy-poor countries becomes significant as more mature aged students are encouraged to attend schools even at night, providing greater flexibility

Table 7

Threshold effect in the relationship between Education and Energy Poverty.

Dependent Variable →	Thresholds in Poverty Headcount ratio		Thresholds in Income Per Capita	
	Progression to secondary school	Average years of Schooling	Progression to secondary school	Progression to secondary school
<i>Test for Threshold effect</i>				
Number of Significant Thresholds	0	0	0	0
F-Statistics	16.66	13.96	45.02	40.31
P-Value	0.907	0.956	0.460	0.637

Notes: (1) The values in square brackets are 95% confidence intervals (2) The Standard errors in parentheses. (3) ***, **, and * denote significant at the 1%, 5% and 10% level respectively. (4) As none of the specifications in the above model had any significant thresholds detected, the results boil down to fixed effect model presented in [Tables 4 & 5](#). Country fixed effects are included in all models.

to working conditions ([Toman and Jemelkova, 2003](#)).

4.4. Threshold estimates

This section empirically tests the conditional effect of the poverty headcount ratio and income per capita in the relationship between energy poverty and development outcomes in the health and education sectors. [Table 6](#) presents the threshold regression estimates of health variables with the EDI. The first two columns represent the specifications where the poverty headcount ratio is used as a threshold variable. In contrast, the last two columns represent the specifications where GDP per capita is used as the threshold variable. We sequentially conduct the tests for one, two and three thresholds to determine the number of endogenous thresholds in each model. If the test for one threshold model is not significant, we conclude that the relationship does not have a threshold effect; however, if we find a single threshold to be significant, we carry on with testing for two thresholds. If the two-threshold test is significant, we carry on with testing for three thresholds and so on. If the two-threshold test is insignificant, we conclude that the relationship has only one significant threshold. If the two-threshold test is significant, but the three-threshold test is insignificant, we conclude that the model has two endogenous thresholds. The [Hansen \(1999\)](#) test has a limitation that it allows for a maximum of three thresholds only; however, as none of our specifications has more than two thresholds, this limitation was inconsequential.

Looking at the threshold effect of poverty on health outcomes, we find two significant poverty thresholds (at 31.4% and 49.2%) in the relationship between the EDI and life expectancy, whereas there is one significant threshold in the relationship between the EDI and infant mortality. Focussing on the relationship between the EDI and life expectancy, we note that at the higher levels of poverty head count, the marginal contribution of the EDI in improving life expectancy increases. More specifically, at the poverty head count level of 31.4 or below one standard deviation (SD) increase in the EDI is associated with a 0.33 SD increase in life expectancy. Similarly, at poverty head count level in the range of 31.4 to 49.2, a one SD increase in the EDI is associated with a 0.50 (around 1.5 times) SD increase in life expectancy. For poverty rates above 49.2, one SD increase in the EDI is associated with a 0.89 SD (about 2.7 times compared to lowest poverty case) increase in life expectancy. The finding suggests that the marginal benefits of reducing energy poverty are highest for the poorest countries in terms of the gains associated with increased life expectancy. This result could also indicate that the countries with the highest levels of poverty have a very low baseline for life expectancy; hence improvements in the EDI are associated with relatively significant life expectancy gains.

While considering the other health measure of infant mortality, we

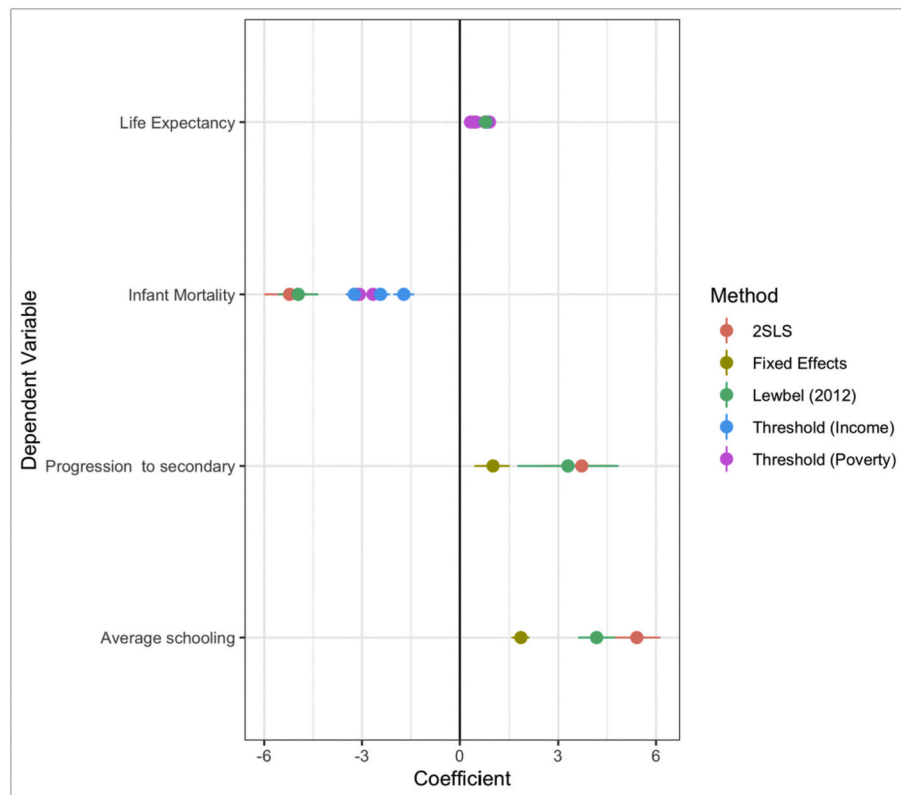


Fig. 5. Coefficients plots showing the relationship between health and education outcomes with energy development index.

note that marginal gains of the EDI in terms of reducing infant mortality diminish as poverty increases. We only find one significant poverty threshold in this relationship. For poverty less than 5.80, we find that a one SD increase in the EDI reduces the infant mortality by -3.08 SD. Conversely, when poverty head count is higher than 5.80, we find that the impact of one SD rise in the EDI is associated with only a -2.67 SD decline in infant mortality.

Next, when income per capita is employed as a threshold variable, we do not find any significant threshold for the relationship between the EDI and life expectancy, whereas we detect two significant thresholds for income per capita in the relationship between the EDI and infant mortality rate. Thus, as income per capita increases, the marginal gains associated with an increase in the EDI reduce infant mortality. Specifically, at the GDP per capita level of around 1600 USD (2010 constant prices) or less, a one SD increase in the EDI is associated with around -1.72 SD decline in infant mortality, whereas in the GDP per capita range from 1600 to 3200 USD (in 2010 constant prices) a one SD increase in the EDI is associated with a -2.43 SD decline (around 1.4 times) in infant mortality. For GDP per capita of above 3200 USD (in 2010 constant prices), a one SD increase in the EDI is associated with a -3.24 SD drop in infant mortality (around 1.8 times compared to lowest income group).

After interpreting the results of Table 6, we can conclude an asymmetric relationship between the EDI and health depending upon the choice of health measure (life expectancy or infant mortality) and the choice of threshold variable (poverty or income per capita). In the relationship between the EDI and life expectancy, we find either there is no threshold (for GDP per capita as a threshold variable), or the marginal benefit of an increased EDI accrues progressively to the most impoverished nations (poverty as a threshold variable). On the other hand, looking at the relationship between the EDI and infant mortality, we notice that the increased EDI's marginal benefits go mostly to the least poor countries or highest per capita income countries when it comes to reducing infant mortality.

Table 7 presents the results of the threshold regression for the relationship between the EDI and the educational outcomes. We do not find any significant thresholds in either using poverty or income per capita as a threshold variable. As none of the endogenous thresholds is significant, the results, in this case, boil down to the fixed effect, 2SLS and Lewbel estimates presented earlier in Tables 4 and 5, respectively.

A possible explanation for the asymmetry in the results obtained in Tables 6 and 7, could be that specific education outcomes largely depend on the availability of educational infrastructure, and energy poverty impacts these educational outcomes by altering the effectiveness of that infrastructure. The UN records that primary education enrolment in developing countries has reached more than 90% by 2010. Further, the proportion of children who dropped out of primary and secondary education drastically declined from 26% in 2000 to only 17% in 2018.⁵ Whilst universal public schooling is hitherto established in many developing countries, education outcomes, such as progression to secondary schooling and average years of schooling, depend less on the affordability aspect of poverty and income per capita, and more on the access and availability of schools. This probably explains why we do not find any thresholds in poverty or per capita income for the EDI and education regressions. However, there is a direct positive effect of lowering energy poverty on educational outcomes by altering the effectiveness of the public provision of education infrastructure, such as better learning environments, evening classes and providing the opportunity for women and children to attend school instead of spending time collecting firewood and traditional cooking fuels. This is supported by our earlier findings in Tables 4 and 5, respectively.

The health outcomes in developing countries, on the other hand, are more likely to be affected by the affordability aspect of households. No matter how much a government invests in the provision of public health infrastructure, the health outcomes, such as life expectancy and infant

⁵ See <https://www.un.org/sustainabledevelopment/education/>

mortality, largely depend on the living standards and how much a household can afford. The affordability factors could be the quality of the home, choice of cooking fuel, income status and the nutritional content of food consumed. Whilst the consumption of everyday goods and services are subject to diminishing marginal utility, adding extra years of life to consumption does not fall under diminishing marginal utility, regardless whether the household is poor or rich (Hall and Jones, 2007). Thus, with increasing health care costs and the absence of a good quality public healthcare system in developing countries, there is an additional burden on poor households to live longer and compete with the more affluent section of the population in accessing healthcare services (O'Donnell et al., 2015). Hence, poverty or per capita income has a direct threshold effect on this relationship between the EDI and health. In terms of policy implications, we can conclude that any changes to energy policies aimed at improving the health outcomes, should be calibrated by taking into account the differential nature of the relationship between the EDI and health, based on the endogenous threshold effect of poverty and income per capita. Whereas energy policies aimed at improving the educational outcomes need not make any adjustments for threshold effects and can be expected to have a uniform impact across the board.

The effect of the EDI on health and education outcomes under various empirical methods can be summarised using Fig. 5. Fig. 5 plots the coefficients of the EDI for all different empirical methods applied in Tables 2–7, namely, OLS panel fixed effects, FE-2SLS, Lewbel estimates, endogenous threshold regression models with poverty headcount ratio as the threshold variable, and finally, endogenous threshold regression models with income per capita as the threshold variable. We find that the magnitude of the positive coefficient of the EDI on life expectancy is robust and close to each other across all five empirical models (between 0 and 1.5). In contrast, the negative coefficient on infant mortality varies significantly across all five models with the maximum effect observed in FE-2SLS and Lewbel estimations (greater than -4.5 compared to close to -3 or below in other estimation methods). We find similar evidence for both of the educational outcomes. The magnitude of the EDI coefficient increases under FE-2SLS and Lewbel estimations, which control for the endogeneity relationship between the EDI and education outcomes. Thus, empirically the coefficients across different estimation methods are robust with the maximum effect of the EDI realised after we control for endogeneity in the regression estimates.

5. Conclusion

Until now, the literature on energy poverty and its relationship with health and education outcomes are primarily confined to a single country or a small group of countries using household-level data. International evidence on access to electricity and energy use and their effects on the health and education outcomes are nearly absent from the extant literature. This paper fills this gap and considers a large sample of 50 developing countries from Asia, Africa, Latin America and Europe in the period 1990–2017. Previous research conjectures that there is an increasing returns to scale effect on economic development and productivity increasing from positive education and health-related benefits due to higher energy efficiency (Toman and Jemelkova, 2003). Our findings support this hypothesis empirically at an international level and show that greater energy development is associated with better quality health and education outcomes. More specifically, our results show that higher energy development leads to higher life expectancy rates, lower infant mortality rates, a higher progression from primary to secondary schooling and higher average years of schooling. These results are robust to adopting alternative estimation methods and after controlling for endogeneity. Comparing the key dimensions of energy development, we find that access to electricity has a higher and significant positive effect on development outcomes than energy use. This finding is critical to the sustainable development of poor and emerging economies where electricity access is still a barrier in the twenty-first century, particularly in

Sub-Saharan Africa and South Asian countries as stipulated by the World Bank in their 2018 report.⁶ Therefore, findings of this study would help policymakers draw common policy conclusions applicable to the entire developing world which aims to achieve the SDG target of alleviating energy poverty worldwide by 2030.

Another uniqueness of this study is the construction of an EDI of all 50 countries on an annual basis following the IEA (2010) report on energy poverty and the UNDP's method of constructing the human development index. The study adopts the methodology of creating a multidimensional energy poverty index similar to the micro-level studies in the extant literature using household level data and applying that to a macro-level analysis suitable for cross country comparison of developing countries. The aggregate index comprises four key indicators of energy development. They are electricity access, electricity usage, renewable energy use and total energy consumption. Thus, the higher the value of the EDI, the lower the level of energy poverty. By comparing the EDI constructed in this study, energy progression can be tracked over time at an international level. This will also help policy-makers implement policies, such as more targeted foreign aid disbursements in the energy sector to increase energy security and efficiency (Maruta and Banerjee, 2020).

As a final contribution, the paper considers poverty headcount ratio and income per capita as possible threshold channels through which energy poverty can influence the health and education sectors. Both variables have been identified in the literature as potential determinants of health and education outcomes; however, to date there is no empirical evidence that the relationship between energy poverty and development outcomes is conditioned by these variables. Our results show that poverty and income levels have a significant threshold effect in the health sector; however, a similar effect is absent in the education sector. This is an interesting finding given the literature largely treats the health and education sectors as equally important and draws common policy conclusions for both. Moreover, the poverty headcount ratio's threshold effect is different for life expectancy rate compared to infant mortality rate. The marginal gain from lower energy poverty increases life expectancy rates where there is a higher threshold level of poverty, indicating that energy development has the maximum effect when poverty is paramount. In contrast, energy development is more beneficial for reducing the infant mortality rate when the poverty level is low, or income per capita is high. This further suggests that the relationship between energy and the infant mortality rate is very much dependent on affordable healthcare services or access to better quality cooking fuels or environmental conditions.

Our empirical findings also suggest that affordability or lack of access to better public services is a more significant barrier in the health sector than the education sector. This may be because the goal of achieving universal primary education is already successful to a large extent in many developing countries.⁷ Although developing countries have to go a long way forward to close the gap between low-and high-quality education services to the poor, with universal primary education systems widely implemented across many developing countries, the threshold effect of poverty and income remains a lesser challenge in education than the health sector. In the absence of well-developed universal public health systems, there is a greater prevalence of health inequality among population segments (O'Donnell et al., 2015). Thus, lack of access to affordable and quality healthcare is more pronounced for poorer households who experience higher financial burdens to compete with the wealthier sections of the population. Hence, policy-makers should aim to increase energy efficacy in poorer segments of the

⁶ See <https://www.worldbank.org/en/news/feature/2018/04/18/access-energy-sustainable-development-goal-7>

⁷ According to the United Nation's 2015 report, the enrolment in primary education in the developing regions has gone up to 91% from 83% in 2000. See <https://www.un.org/millenniumgoals/education.shtml>

population and reduce the general poverty level worldwide, further enhancing the positive effect of energy development on health outcomes.

Credit author statement

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Admasu Asfaw Maruta: Data Collection, Software, Data Analysis, Methodology, Writing - Review & Editing.

Appendix 1. List of countries

Angola	Ghana	Paraguay	Zambia
Argentina	India	Peru	Zimbabwe
Azerbaijan	Indonesia	Philippines	
Bangladesh	Iran	Senegal	
Benin	Iraq	South Africa	
Bolivia	Jordan	Sri Lanka	
Botswana	Kenya	Sudan	
Brazil	Kyrgyz Republic	Syria	
Cameroon	Malaysia	Tanzania	
Chile	Morocco	Thailand	
Colombia	Mozambique	Togo	
Cote D'Ivoire	Namibia	Tunisia	
Ecuador	Nepal	Uruguay	
Egypt	Niger	Venezuela	
Ethiopia	Nigeria	Vietnam	
Gabon	Pakistan	Yemen	

Appendix 2. Definition and sources of data

Variable	Description	Source
Life expectancy at birth, total (years)	It indicates the number of years a newborn infant would live if prevailing patterns of mortality at the time of its birth were to stay the same throughout its life.	WDI
Infant mortality rate (per 1000 live births)	Infant mortality rate is the number of infants dying before reaching one year of age, per 1000 live births in a given year.	WDI
Progression to secondary school (%)	Progression to secondary school refers to the number of new entrants to the first grade of secondary school in a given year as a percentage of the number of students enrolled in the final grade of primary school in the previous year (minus the number of repeaters from the last grade of primary education in the given year).	WDI
Average (mean) years of schooling	Average number of completed years of education of a country's population aged 15 years and older, excluding years spent repeating individual grades.	Barro-Lee database
Energy Development Index	Geometric mean of Energy use (kilo of oil equivalent per capita), renewable energy consumption (% of total final energy consumption), access to electricity (% of population) and electric power consumption (kWh per capita)	WDI
Poverty headcount ratio at \$1.90 a day (2011 PPP) (% of population)	Poverty headcount ratio at \$1.90 a day is the percentage of the population living on less than \$1.90 a day at 2011 international prices. For missing figures, the series is spliced with Poverty headcount ratio at national poverty lines (% of population).	WDI
Real GDP per capita (or GDP per capita (constant 2010 US\$))	GDP per capita is gross domestic product divided by midyear population. GDP is the sum of gross value added by all resident producers in the economy plus any product taxes and minus any subsidies not included in the value of the products. It is calculated without making deductions for depreciation of fabricated assets or for depletion and degradation of natural resources. Data are in constant 2010 U.S. dollars.	WDI
Access to immunization, measles (% of children ages 12–23 months)	Child immunization, measles, measures the percentage of children ages 12–23 months who received the measles vaccination before 12 months or at any time before the survey. A child is considered adequately immunized against measles after receiving one dose of vaccine.	WDI
Pupil-teacher ratio, primary	Primary school pupil-teacher ratio is the average number of pupils per teacher in primary school.	WDI
Prevalence of undernourishment	Population below minimum level of dietary energy consumption shows the percentage of the population whose food intake is insufficient to meet dietary energy requirements continuously.	WDI
Rule of law	Reflects perceptions of the extent to which agents have confidence in and abide by the rules of society, and in particular the quality of contract enforcement, property rights, the police, and the courts, as well as the likelihood of crime and violence. Estimate of governance (ranges from approximately –2.5 (weak) to 2.5 (strong)).	Worldwide Governance Indicators
Urban population (% of total population)	Urban population refers to people living in urban areas as defined by national statistical offices. It is calculated using World Bank population estimates and urban ratios from the United Nations World Urbanization Prospects. The data is collected and smoothed by United Nations Population Division.	WDI
Night-time luminosity (values 0–63)	Yearly night-time luminosity data is compiled by Hodler and Raschky (2014) and available from the website https://datainspace.org . The data is available from 1990 to 2013. The data between 2014 and 2017 is further extrapolated to construct a complete series on annual basis in the period 1990–2017.	Hodler and Raschky (2014)

Appendix 3. Correlation table

	Life expectancy at birth, total (years)	Infant mortality rate	Progression to secondary school	Average years of schooling	Energy Development Index	Access to electricity	Energy use	Rule of law	Real GDP per capita	Poverty headcount ratio	Urban population growth	Prevalence of undernourishment	Access to immunization	Pupil-teacher ratio, primary
Life expectancy at birth, total (years)	1.00													
Infant mortality rate	−0.87	1.00												
Progression to secondary school	0.59	−0.65	1.00											
Average years of schooling	0.55	−0.67	0.42	1.00										
Energy Development Index	0.48	−0.59	0.34	0.67	1.00									
Access to electricity	0.86	−0.77	0.50	0.65	0.62	1.00								
Energy use	0.36	−0.45	0.28	0.55	0.69	0.52	1.00							
Rule of law	0.33	−0.39	0.29	0.29	0.39	0.20	0.18	1.00						
Real GDP per capita	0.51	−0.57	0.26	0.53	0.80	0.57	0.73	0.31	1.00					
Poverty headcount ratio	−0.78	0.75	−0.53	−0.59	−0.51	−0.85	−0.44	−0.18	−0.52	1.00				
Urban population growth	−0.52	0.52	−0.28	−0.54	−0.48	−0.58	−0.28	−0.12	−0.38	0.52	1.00			
Prevalence of undernourishment	−0.62	0.61	−0.43	−0.38	−0.46	−0.61	−0.34	−0.37	−0.45	0.53	0.41	1.00		
Access to immunization	0.67	−0.75	0.46	0.57	0.43	0.55	0.27	0.39	0.33	−0.55	−0.44	−0.44	1.00	
Pupil-teacher ratio, primary	−0.64	0.63	−0.47	−0.56	−0.44	−0.71	−0.43	−0.19	−0.46	0.71	0.37	0.41	−0.48	1.00

Appendix 4. Country-level summary statistics, 1990–2017

No	Country	Years	Energy Variables			Health Variables				Education variables			Poverty headcount ratio
			Energy Development Index	Access to electricity	Energy use	Life expectancy at birth, total (years)	Infant mortality rate	Prevalence of undernourishment	Access to immunization	Progression to secondary school	Average years of schooling	Pupil-teacher ratio, primary	
1	Angola	28	0.15	27.64	485.45	50.70	100.82	61.51	48.61	30.15	4.57	41.11	35.53
2	Argentina	28	0.39	97.10	1738.12	74.12	16.41	2.40	94.68	94.22	9.39	19.20	4.02
3	Azerbaijan	28	0.26	98.70	1669.17	68.37	50.32	18.20	74.25	95.48	10.38	15.71	3.36
4	Bangladesh	28	0.13	41.72	170.05	66.40	57.35	16.86	81.75	84.25	4.41	48.37	28.02
5	Benin	28	0.12	25.88	355.09	57.23	81.38	17.10	67.75	76.57	2.63	46.78	48.09
6	Bolivia	28	0.23	75.52	577.51	64.16	49.21	26.76	79.29	92.26	7.49	22.37	14.69
7	Botswana	28	0.31	34.76	1079.70	57.13	30.40	23.41	92.57	95.41	7.86	26.09	24.49
8	Brazil	28	0.50	95.79	1184.35	71.24	27.61	10.99	94.00	90.03	5.98	25.16	10.19
9	Cameroon	28	0.20	45.64	376.92	53.57	77.21	21.54	60.25	47.91	4.94	48.81	39.05
10	Chile	28	0.53	97.65	1658.34	77.08	9.18	3.86	94.14	97.89	9.26	29.24	2.79
11	Colombia	28	0.33	94.85	695.17	73.61	19.66	8.55	89.14	93.48	6.81	26.60	10.84
12	Cote D'Ivoire	28	0.21	51.18	470.45	52.20	86.50	18.78	65.14	70.12	3.54	42.12	26.94
13	Ecuador	28	0.27	94.61	726.51	73.33	23.07	15.22	88.29	79.61	7.50	24.90	12.59
14	Egypt	28	0.24	97.74	725.33	68.82	35.05	4.91	93.96	93.42	5.39	24.74	4.19
15	Ethiopia	28	0.11	25.20	483.56	55.83	76.33	44.83	45.18	93.40	1.73	52.96	51.13
16	Gabon	28	0.51	80.43	1931.56	60.64	48.54	10.49	61.61	71.28	6.54	36.29	8.58
17	Ghana	28	0.20	52.15	331.51	59.17	57.78	14.84	81.14	95.87	6.23	31.27	27.57
18	India	28	0.25	65.88	473.19	63.85	59.36	14.19	68.21	91.36	4.66	38.68	36.63
19	Indonesia	28	0.29	83.17	752.48	66.96	37.96	16.56	75.89	81.79	6.38	20.21	28.54
20	Iran	28	0.21	98.55	2176.29	71.26	25.28	4.24	96.82	94.05	7.27	26.56	2.16
21	Iraq	28	0.16	98.16	1276.03	68.61	32.81	22.59	75.61	79.96	5.40	17.48	1.38
22	Jordan	28	0.21	98.88	971.53	72.28	21.03	13.15	94.04	98.80	9.24	21.05	1.01
23	Kenya	28	0.17	24.06	764.59	57.00	51.72	25.63	83.50	94.00	5.43	31.20	37.09
24	Kyrgyz Republic	28	0.35	99.71	670.17	68.51	36.94	16.25	97.21	96.97	9.94	23.37	11.15
25	Malaysia	28	0.39	99.45	2288.20	73.32	8.80	2.39	92.68	96.35	8.57	16.63	0.55
26	Morocco	28	0.21	76.65	444.36	70.80	37.90	6.59	93.00	87.81	3.72	27.18	3.46
27	Mozambique	28	0.15	12.64	421.10	50.62	102.01	35.46	73.89	60.41	2.36	59.47	74.79
28	Namibia	28	0.28	38.32	649.96	56.69	41.23	15.60	71.61	93.48	6.01	30.32	35.91
29	Nepal	28	0.15	44.93	354.34	63.63	54.74	20.87	74.79	85.69	2.98	34.66	42.68
30	Niger	28	0.06	9.97	132.61	52.57	86.34	20.82	50.32	57.39	1.25	39.78	67.89
31	Nigeria	28	0.20	45.83	733.27	48.57	100.85	8.82	43.50	45.72	5.40	35.75	58.38
32	Pakistan	28	0.24	70.56	451.64	63.63	80.72	16.88	58.96	70.10	3.87	39.16	20.14
33	Paraguay	28	0.40	90.75	753.58	71.30	26.05	12.39	84.86	87.85	6.90	24.39	6.26
34	Peru	28	0.28	78.86	535.22	71.98	27.44	12.77	88.14	94.71	8.10	23.71	13.50
35	Philippines	28	0.25	78.49	464.29	69.04	28.37	17.19	82.86	97.46	8.14	33.12	12.23
36	Senegal	28	0.16	43.11	489.61	60.95	55.86	20.99	69.71	63.17	2.35	44.70	46.87
37	South Africa	28	0.54	74.02	2587.72	58.84	40.39	4.48	75.39	97.17	8.82	33.42	26.00
38	Sri Lanka	28	0.25	76.02	431.14	73.01	12.98	15.06	94.46	98.30	10.06	25.46	5.91
39	Sudan	28	0.14	34.59	384.25	60.06	61.26	21.47	68.39	96.80	2.67	28.24	19.55
40	Syria	28	0.17	89.36	880.48	72.17	20.11	13.27	78.21	70.56	5.20	30.05	1.44
41	Tanzania	28	0.11	12.94	429.87	54.69	67.58	28.64	86.14	35.57	4.68	44.83	66.94
42	Thailand	28	0.42	90.77	1404.94	72.40	17.08	16.00	93.36	90.25	6.41	18.26	1.85
43	Togo	28	0.15	26.68	418.33	56.05	68.64	28.13	63.64	79.24	4.03	44.14	56.74
44	Tunisia	28	0.28	95.53	798.43	73.47	23.91	3.68	93.75	90.40	5.46	21.21	5.31
45	Uruguay	28	0.49	98.55	1019.36	75.35	13.18	3.50	94.25	73.21	7.99	17.64	0.44
46	Venezuela	28	0.47	98.82	2234.47	72.22	18.47	11.59	80.39	96.68	7.37	22.13	10.61
47	Vietnam	28	0.27	89.76	464.30	73.47	23.16	19.40	94.68	96.07	6.16	26.06	25.64
48	Yemen	28	0.07	55.98	285.07	62.36	60.96	29.58	61.96	86.90	1.73	24.47	10.45
49	Zambia	28	0.26	21.70	628.20	50.59	74.75	45.70	87.07	50.24	6.17	47.35	55.83
50	Zimbabwe	28	0.31	34.97	830.41	50.53	51.15	42.02	81.61	75.74	6.65	37.91	30.57

Appendix 5. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eneco.2021.105447>.

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